Decoder-free reproductions of magic-state simulations with real non-Clifford gates on a GPU

Generated by `repro/make_report.py`

2026-11-05 11:30

Every figure below is regenerated from scratch by `uv run python -m repro.run_all`: circuits come from the authors' generators, are sampled with the Tsim CUDA statevector executor (postselection on the device), and every point records its own GPU time, compile time, shot count, GPU model and Tsim build. Where the paper simulated a Clifford proxy (S in place of T, or a $|Y\rangle$ / $|+i\rangle$ state), the same proxy is sampled again on the GPU and plotted next to the published numbers, as a check of the GPU kernel; the real non-Clifford circuit (T gates, rotations) is sampled alongside. Logical error rates are per kept shot; bands and error bars span hypotheses within a factor 1000 of the maximum likelihood (the convention of `sinter`). Triangles mark upper bounds for points with zero observed errors.

Tsim builds: a2af975fbc.

study	points	GPU sampling	shots
gidney_cultivation	35	28.8 min	7.5e12
wan_few_clifford_terms	16	4.5 min	6.5e11
soft_d5	5	7.9 min	2.8e12
clifft_inject_cultivate	0	0.0 s	0
gauging_cultivation	54	33.2 min	9.3e12
fold_transversal	12	46.4 min	3.7e12
chan_flags	32	41.3 min	1.1e13
magna	28	11.7 min	3e12
google_experiment	278	4.4 min	1.5e12
star_injection	30	20.4 min	7.6e12
rp2_ungrown	2	23 s	7.9e10
erasure_injection	6	17.0 min	2.2e12
vaknin_full_vector	0	0.0 s	0

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Reproduction checks against published numbers

Every published number (tables, released data) that a sampled point can be compared with. ±: one standard deviation (Poisson for error counts, binomial for acceptance; published values without an uncertainty, or exact values, have none). z: difference over the combined standard deviation.

study	point	variant	quantity	published	Tsim GPU	ratio
gidney_cultivation	bell d=3 p=0.0005 S (Stim)	clifford	LER	4.91e-08 ± 2e-09	4.89e-08 ± 3e-10	0.996
gidney_cultivation	bell d=3 p=0.0005 S (Stim)	clifford	acceptance	0.846 ± 2e-06	0.846 ± 5e-07	1.000
gidney_cultivation	bell d=3 p=0.001 S (Stim)	clifford	LER	3.92e-07 ± 1e-08	3.99e-07 ± 3e-09	1.019
gidney_cultivation	bell d=3 p=0.001 S (Stim)	clifford	acceptance	0.716 ± 7e-06	0.716 ± 2e-06	1.000
gidney_cultivation	bell d=3 p=0.002 S (Stim)	clifford	LER	3.25e-06 ± 1e-07	3.32e-06 ± 2e-08	1.020
gidney_cultivation	bell d=3 p=0.002 S (Stim)	clifford	acceptance	0.513 ± 2e-05	0.513 ± 5e-06	1.000
gidney_cultivation	teleport d=3 p=0.0005 S (Stim)	clifford	LER	7.28e-08 ± 2e-09	7.12e-08 ± 5e-10	0.978
gidney_cultivation	teleport d=3 p=0.0005 S (Stim)	clifford	acceptance	0.804 ± 3e-06	0.804 ± 7e-07	1.000
gidney_cultivation	teleport d=3 p=0.001 S (Stim)	clifford	LER	5.79e-07 ± 2e-08	5.89e-07 ± 4e-09	1.018
gidney_cultivation	teleport d=3 p=0.001 S (Stim)	clifford	acceptance	0.646 ± 9e-06	0.646 ± 2e-06	1.000
gidney_cultivation	teleport d=3 p=0.002 S (Stim)	clifford	LER	4.81e-06 ± 2e-07	4.83e-06 ± 3e-08	1.005
gidney_cultivation	teleport d=3 p=0.002 S (Stim)	clifford	acceptance	0.418 ± 2e-05	0.418 ± 5e-06	1.000
gidney_cultivation	unitary d=3 p=0.0005 S (Stim)	clifford	LER	4e-08 ± 1e-09	4.11e-08 ± 3e-10	1.027
gidney_cultivation	unitary d=3 p=0.0005 S (Stim)	clifford	acceptance	0.829 ± 2e-06	0.829 ± 5e-07	1.000
gidney_cultivation	unitary d=3 p=0.001 S (Stim)	clifford	LER	3.42e-07 ± 1e-08	3.43e-07 ± 2e-09	1.001
gidney_cultivation	unitary d=3 p=0.001 S (Stim)	clifford	acceptance	0.687 ± 7e-06	0.687 ± 2e-06	1.000
gidney_cultivation	unitary d=3 p=0.001 T (state vector)	t	LER	9.73e-07 ± 5e-08	8.98e-07 ± 6e-09	0.923
gidney_cultivation	unitary d=3 p=0.001 T (state vector)	t	acceptance	0.687 ± 2e-05	0.687 ± 3e-06	1.000
gidney_cultivation	unitary d=3 p=0.001 S (state vector)	clifford	LER	3.89e-07 ± 3e-08	3.43e-07 ± 2e-09	0.880
gidney_cultivation	unitary d=3 p=0.001 S (state vector)	clifford	acceptance	0.687 ± 2e-05	0.687 ± 2e-06	1.000
gidney_cultivation	unitary d=3 p=0.001 Z (state vector)	clifford	LER	2.93e-07 ± 3e-08	2.88e-07 ± 2e-09	0.983
gidney_cultivation	unitary d=3 p=0.001 Z (state vector)	clifford	acceptance	0.687 ± 2e-05	0.687 ± 1e-06	1.000
gidney_cultivation	unitary d=3 p=0.002 S (Stim)	clifford	LER	2.85e-06 ± 9e-08	2.95e-06 ± 2e-08	1.037
gidney_cultivation	unitary d=3 p=0.002 S (Stim)	clifford	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1.000
gidney_cultivation	unitary d=3 p=0.002 T (state vector)	t	LER	5.86e-06 ± 1e-07	5.69e-06 ± 4e-08	0.972
gidney_cultivation	unitary d=3 p=0.002 T (state vector)	t	acceptance	0.472 ± 2e-05	0.472 ± 6e-06	1.000
gidney_cultivation	unitary d=3 p=0.002 S (state vector)	clifford	LER	2.87e-06 ± 1e-07	2.95e-06 ± 2e-08	1.028
gidney_cultivation	unitary d=3 p=0.002 S (state vector)	clifford	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1.000
gidney_cultivation	unitary d=3 p=0.002 Z (state vector)	clifford	LER	2.53e-06 ± 1e-07	2.46e-06 ± 2e-08	0.973
gidney_cultivation	unitary d=3 p=0.002 Z (state vector)	clifford	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1.000
gidney_cultivation	unitary d=3 p=0.003 T (state vector)	t	LER	1.81e-05 ± 6e-07	1.81e-05 ± 1e-07	1.000
gidney_cultivation	unitary d=3 p=0.003 T (state vector)	t	acceptance	0.325 ± 4e-05	0.325 ± 8e-06	1.000
gidney_cultivation	unitary d=3 p=0.003 S (state vector)	clifford	LER	1.03e-05 ± 4e-07	1.05e-05 ± 7e-08	1.027
gidney_cultivation	unitary d=3 p=0.003 S (state vector)	clifford	acceptance	0.325 ± 3e-05	0.325 ± 6e-06	1.000
gidney_cultivation	unitary d=3 p=0.003 Z (state vector)	clifford	LER	9.28e-06 ± 4e-07	9e-06 ± 6e-08	0.970
gidney_cultivation	unitary d=3 p=0.003 Z (state vector)	clifford	acceptance	0.325 ± 3e-05	0.325 ± 6e-06	1.000
gidney_cultivation	unitary d=3 p=0.005 T (state vector)	t	LER	8.75e-05 ± 3e-06	8.48e-05 ± 4e-07	0.969
gidney_cultivation	unitary d=3 p=0.005 T (state vector)	t	acceptance	0.155 ± 4e-05	0.155 ± 6e-06	0.999
gidney_cultivation	unitary d=3 p=0.005 S (state vector)	clifford	LER	5.81e-05 ± 2e-06	5.69e-05 ± 3e-07	0.980
gidney_cultivation	unitary d=3 p=0.005 S (state vector)	clifford	acceptance	0.155 ± 3e-05	0.155 ± 6e-06	1.000
gidney_cultivation	unitary d=3 p=0.005 Z (state vector)	clifford	LER	4.92e-05 ± 2e-06	4.84e-05 ± 3e-07	0.983
gidney_cultivation	unitary d=3 p=0.005 Z (state vector)	clifford	acceptance	0.155 ± 3e-05	0.155 ± 6e-06	1.000
gidney_cultivation	unitary d=3 p=0.007 T (state vector)	t	LER	0.00026 ± 8e-06	0.000252 ± 2e-06	0.970
gidney_cultivation	unitary d=3 p=0.007 T (state vector)	t	acceptance	0.0741 ± 4e-05	0.0741 ± 7e-06	1.001
gidney_cultivation	unitary d=3 p=0.007 S (state vector)	clifford	LER	0.00019 ± 6e-06	0.000184 ± 7e-07	0.966
gidney_cultivation	unitary d=3 p=0.007 S (state vector)	clifford	acceptance	0.0741 ± 3e-05	0.0742 ± 3e-06	1.000
gidney_cultivation	unitary d=3 p=0.007 Z (state vector)	clifford	LER	0.000154 ± 5e-06	0.000154 ± 6e-07	1.003
gidney_cultivation	unitary d=3 p=0.007 Z (state vector)	clifford	acceptance	0.0743 ± 3e-05	0.0743 ± 3e-06	1.000
gidney_cultivation	unitary d=3 p=0.01 T (state vector)	t	LER	0.00091 ± 3e-05	0.000866 ± 5e-06	0.951
gidney_cultivation	unitary d=3 p=0.01 T (state vector)	t	acceptance	0.0249 ± 2e-05	0.0249 ± 4e-06	0.998
gidney_cultivation	unitary d=3 p=0.01 S (state vector)	clifford	LER	0.000667 ± 2e-05	0.000685 ± 4e-06	1.028
gidney_cultivation	unitary d=3 p=0.01 S (state vector)	clifford	acceptance	0.0249 ± 2e-05	0.0249 ± 4e-06	1.001
gidney_cultivation	unitary d=3 p=0.01 Z (state vector)	clifford	LER	0.000573 ± 2e-05	0.000585 ± 4e-06	1.021
gidney_cultivation	unitary d=3 p=0.01 Z (state vector)	clifford	acceptance	0.025 ± 2e-05	0.025 ± 4e-06	1.000

study	point	variant	quantity	published	Tsim GPU	ratio
gidney_cultivation	unitary d=5 p=0.0005 S (Stim)	clifford	acceptance	0.379 ± 5e-07	0.379 ± 1e-06	1.000
gidney_cultivation	unitary d=5 p=0.002 S (Stim)	clifford	LER	1.26e-08 ± 8e-10	1.05e-08 ± 4e-10	0.831
gidney_cultivation	unitary d=5 p=0.002 S (Stim)	clifford	acceptance	0.0208 ± 1e-07	0.0208 ± 7e-08	1.000
wan_few_clifford_terms	d=3 p=0.0005 T	t	LER	1.8e-07 ± 7e-09	1.83e-07 ± 1e-09	1.022
wan_few_clifford_terms	d=3 p=0.0005 T	t	acceptance	0.807 ± 6e-06	0.804 ± 1e-06	0.997
wan_few_clifford_terms	d=3 p=0.001 T	t	LER	1e-06 ± 5e-08	1.06e-06 ± 7e-09	1.055
wan_few_clifford_terms	d=3 p=0.001 T	t	acceptance	0.651 ± 2e-05	0.647 ± 3e-06	0.994
wan_few_clifford_terms	d=3 p=0.0015 T	t	LER	2.95e-06 ± 1e-07	3.09e-06 ± 2e-08	1.049
wan_few_clifford_terms	d=3 p=0.0015 T	t	acceptance	0.525 ± 2e-05	0.521 ± 4e-06	0.991
wan_few_clifford_terms	d=3 p=0.002 T	t	LER	6.76e-06 ± 2e-07	6.94e-06 ± 5e-08	1.027
wan_few_clifford_terms	d=3 p=0.002 T	t	acceptance	0.424 ± 2e-05	0.419 ± 6e-06	0.989
wan_few_clifford_terms	d=3 p=0.0025 T	t	LER	1.29e-05 ± 3e-07	1.28e-05 ± 9e-08	0.992
wan_few_clifford_terms	d=3 p=0.0025 T	t	acceptance	0.343 ± 2e-05	0.338 ± 7e-06	0.986
wan_few_clifford_terms	d=3 p=0.003 T	t	LER	2.15e-05 ± 4e-07	2.17e-05 ± 2e-07	1.008
wan_few_clifford_terms	d=3 p=0.003 T	t	acceptance	0.277 ± 2e-05	0.272 ± 8e-06	0.983
wan_few_clifford_terms	d=3 p=0.004 T	t	LER	4.93e-05 ± 7e-07	5.08e-05 ± 3e-07	1.030
wan_few_clifford_terms	d=3 p=0.004 T	t	acceptance	0.181 ± 2e-05	0.177 ± 7e-06	0.978
wan_few_clifford_terms	d=3 p=0.005 T	t	LER	9.77e-05 ± 1e-06	9.9e-05 ± 5e-07	1.012
wan_few_clifford_terms	d=3 p=0.005 T	t	acceptance	0.118 ± 1e-05	0.115 ± 6e-06	0.972
soft_d5	d=5 p=0.002 T (SOFT Table 3)	t	LER	3.67e-08 ± 8e-09	5.33e-08 ± 2e-09	1.455
soft_d5	d=5 p=0.002 T (exact, 2609.18922)	t	LER	5.175e-08	5.33e-08 ± 2e-09	1.031
soft_d5	d=5 p=0.002 T (exact, 2609.18922)	t	acceptance	0.02083	0.0208 ± 1e-07	1.000
clifft_inject_cultivate	d=5 p=0.002 T (Clifft Table 2, importance sampling)	t	LER	5.78e-08	5.07e-08 ± 2e-09	0.877
gauging_cultivation	d=3 p=0.001 |+i> proxy	clifford	LER	8.8e-07 ± 8e-08	8.77e-07 ± 6e-09	0.996
gauging_cultivation	d=3 p=0.001 |+i> proxy	clifford	acceptance	0.46	0.461 ± 2e-06	1.002
gauging_cultivation	d=3 p=0.0005 |+i> proxy	clifford	LER	1.2e-07 ± 2e-08	1.07e-07 ± 8e-10	0.891
gauging_cultivation	d=3 p=0.0005 |+i> proxy	clifford	acceptance	0.65	0.679 ± 9e-07	1.044
gauging_cultivation	d=3 p=0.001 |+i> (notebook)	clifford	LER	8.9e-07 ± 1e-07	8.77e-07 ± 6e-09	0.985
gauging_cultivation	d=3 p=0.001 |+i> (notebook)	clifford	acceptance	0.461 ± 0.0002	0.461 ± 2e-06	1.000
gauging_cultivation	d=3 p=0.002 |+i> (notebook)	clifford	LER	7.62e-06 ± 6e-07	7.58e-06 ± 5e-08	0.995
gauging_cultivation	d=3 p=0.002 |+i> (notebook)	clifford	acceptance	0.213 ± 0.0001	0.213 ± 4e-06	1.000
gauging_cultivation	d=3 p=0.003 |+i> (notebook)	clifford	LER	3.03e-05 ± 2e-06	2.76e-05 ± 2e-07	0.910
gauging_cultivation	d=3 p=0.003 |+i> (notebook)	clifford	acceptance	0.0983 ± 9e-05	0.0983 ± 3e-06	1.000
gauging_cultivation	d=3 p=0.004 |+i> (notebook)	clifford	LER	6.98e-05 ± 4e-06	7.04e-05 ± 4e-07	1.010
gauging_cultivation	d=3 p=0.004 |+i> (notebook)	clifford	acceptance	0.0456 ± 7e-05	0.0456 ± 2e-06	0.999
gauging_cultivation	d=3 p=0.005 |+i> (notebook)	clifford	LER	0.000146 ± 8e-06	0.000148 ± 7e-07	1.011
gauging_cultivation	d=3 p=0.005 |+i> (notebook)	clifford	acceptance	0.0212 ± 5e-05	0.0212 ± 1e-06	0.999
gauging_cultivation	d=3 p=0.001 T (notebook, qiskit)	t	LER	2.19e-07 ± 2e-07	1.49e-06 ± 1e-08	6.816
gauging_cultivation	d=3 p=0.002 T (notebook, qiskit)	t	LER	8.07e-06 ± 2e-06	1.03e-05 ± 7e-08	1.275
gauging_cultivation	d=3 p=0.003 T (notebook, qiskit)	t	LER	2.36e-05 ± 5e-06	3.37e-05 ± 2e-07	1.429
gauging_cultivation	d=3 p=0.004 T (notebook, qiskit)	t	LER	6.2e-05 ± 1e-05	8.24e-05 ± 5e-07	1.329
gauging_cultivation	d=3 p=0.005 T (notebook, qiskit)	t	LER	0.000119 ± 2e-05	0.000168 ± 1e-06	1.409
gauging_cultivation	gauging d=3 p=1e-04 success (pickle)	clifford	acceptance	0.9226	0.907 ± 3e-05	0.983
gauging_cultivation	gauging d=3 p=2e-04 success (pickle)	clifford	acceptance	0.8514	0.823 ± 4e-05	0.967
gauging_cultivation	gauging d=3 p=3e-04 success (pickle)	clifford	acceptance	0.7861	0.747 ± 4e-05	0.950
gauging_cultivation	gauging d=3 p=4e-04 success (pickle)	clifford	acceptance	0.7259	0.678 ± 5e-05	0.934
gauging_cultivation	gauging d=3 p=5e-04 success (pickle)	clifford	acceptance	0.6698	0.615 ± 5e-05	0.919
gauging_cultivation	gauging d=3 p=6e-04 success (pickle)	clifford	acceptance	0.6176	0.558 ± 5e-05	0.904
gauging_cultivation	gauging d=3 p=7e-04 success (pickle)	clifford	acceptance	0.5697	0.507 ± 5e-05	0.889
gauging_cultivation	gauging d=3 p=8e-04 success (pickle)	clifford	acceptance	0.5254	0.46 ± 5e-05	0.875
gauging_cultivation	gauging d=3 p=9e-04 success (pickle)	clifford	acceptance	0.4858	0.417 ± 5e-05	0.859
gauging_cultivation	gauging d=3 p=1e-03 success (pickle)	clifford	acceptance	0.4475	0.379 ± 5e-05	0.846
gauging_cultivation	gauging d=5 p=1e-04 success (pickle)	clifford	acceptance	0.7164	0.674 ± 5e-05	0.941
gauging_cultivation	gauging d=5 p=2e-04 success (pickle)	clifford	acceptance	0.5142	0.454 ± 5e-05	0.883
gauging_cultivation	gauging d=5 p=3e-04 success (pickle)	clifford	acceptance	0.3686	0.306 ± 5e-05	0.830
gauging_cultivation	gauging d=5 p=4e-04 success (pickle)	clifford	acceptance	0.2646	0.206 ± 4e-05	0.780
gauging_cultivation	gauging d=5 p=5e-04 success (pickle)	clifford	acceptance	0.1893	0.139 ± 3e-05	0.735
gauging_cultivation	gauging d=5 p=6e-04 success (pickle)	clifford	acceptance	0.1353	0.0938 ± 3e-05	0.694
gauging_cultivation	gauging d=5 p=7e-04 success (pickle)	clifford	acceptance	0.09691	0.0633 ± 2e-05	0.653
gauging_cultivation	gauging d=5 p=8e-04 success (pickle)	clifford	acceptance	0.06985	0.0427 ± 2e-05	0.611

study	point	variant	quantity	published	Tsim GPU	ratio
gauging_cultivation	gauging d=5 p=9e-04 success (pickle)	clifford	acceptance	0.05023	0.0288 ± 2e-05	0.574
gauging_cultivation	gauging d=5 p=1e-03 success (pickle)	clifford	acceptance	0.03603	0.0194 ± 1e-05	0.539
gauging_cultivation	gauging d=7 p=1e-04 success (pickle)	clifford	acceptance	0.4187	0.358 ± 5e-05	0.854
gauging_cultivation	gauging d=7 p=2e-04 success (pickle)	clifford	acceptance	0.1757	0.128 ± 3e-05	0.728
gauging_cultivation	gauging d=7 p=3e-04 success (pickle)	clifford	acceptance	0.07348	0.0458 ± 2e-05	0.623
gauging_cultivation	gauging d=7 p=4e-04 success (pickle)	clifford	acceptance	0.03091	0.0164 ± 1e-05	0.531
gauging_cultivation	gauging d=7 p=5e-04 success (pickle)	clifford	acceptance	0.01302	0.00589 ± 8e-06	0.453
gauging_cultivation	gauging d=7 p=6e-04 success (pickle)	clifford	acceptance	0.0054	0.00212 ± 5e-06	0.392
gauging_cultivation	gauging d=7 p=7e-04 success (pickle)	clifford	acceptance	0.002271	0.000749 ± 3e-06	0.330
gauging_cultivation	gauging d=7 p=8e-04 success (pickle)	clifford	acceptance	0.000918	0.000272 ± 2e-06	0.296
gauging_cultivation	gauging d=7 p=9e-04 success (pickle)	clifford	acceptance	0.00043	9.67e-05 ± 1e-06	0.225
gauging_cultivation	gauging d=7 p=1e-03 success (pickle)	clifford	acceptance	0.000146	3.38e-05 ± 6e-07	0.232
gauging_cultivation	gauging d=9 p=1e-04 success (pickle)	clifford	acceptance	0.166	0.12 ± 3e-05	0.724
gauging_cultivation	gauging d=9 p=2e-04 success (pickle)	clifford	acceptance	0.02758	0.0144 ± 1e-05	0.523
gauging_cultivation	gauging d=9 p=3e-04 success (pickle)	clifford	acceptance	0.004665	0.00174 ± 4e-06	0.372
gauging_cultivation	gauging d=9 p=4e-04 success (pickle)	clifford	acceptance	0.000764	0.000209 ± 1e-06	0.273
gauging_cultivation	gauging d=9 p=5e-04 success (pickle)	clifford	acceptance	0.000124	2.46e-05 ± 5e-07	0.198
gauging_cultivation	gauging d=9 p=6e-04 success (pickle)	clifford	acceptance	2.4e-05	3.29e-06 ± 2e-07	0.137
gauging_cultivation	gauging d=9 p=7e-04 success (pickle)	clifford	acceptance	1e-06	3e-07 ± 5e-08	0.300
gauging_cultivation	gauging d=9 p=8e-04 success (pickle)	clifford	acceptance	0	2e-08 ± 1e-08	nan
rp2_ungrown	MSC3 S proxy	clifford	LER	8.3e-07	8.25e-07 ± 6e-09	0.995
rp2_ungrown	MSC3 S proxy	clifford	acceptance	0.514	0.514 ± 2e-06	1.000

GPU kernel vs. Stim on the same Clifford circuits

Every Clifford-proxy point re-sampled with Stim on the CPU (`repro/stim_check.py`), same circuit file and postselection: an independent test of the Tsim GPU kernel.

study	point	quantity	Stim (CPU)	Tsim GPU	Stim shots	z
gidney_cultivation	bell_d3_p0.0005_s	acceptance	0.846 ± 1e-05	0.846 ± 5e-07	1e9	-0.5
gidney_cultivation	bell_d3_p0.0005_s	LER	4.96e-08 ± 8e-09	4.89e-08 ± 3e-10	1e9	-0.1
gidney_cultivation	bell_d3_p0.001_s	acceptance	0.716 ± 1e-05	0.716 ± 2e-06	1e9	-0.4
gidney_cultivation	bell_d3_p0.001_s	LER	3.55e-07 ± 2e-08	3.99e-07 ± 3e-09	1e9	+2.0
gidney_cultivation	bell_d3_p0.002_s	acceptance	0.513 ± 2e-05	0.513 ± 5e-06	1e9	-0.2
gidney_cultivation	bell_d3_p0.002_s	LER	3.29e-06 ± 8e-08	3.32e-06 ± 2e-08	1e9	+0.4
gidney_cultivation	teleport_d3_p0.0005_s	acceptance	0.804 ± 1e-05	0.804 ± 7e-07	1e9	+0.2
gidney_cultivation	teleport_d3_p0.0005_s	LER	7.59e-08 ± 1e-08	7.12e-08 ± 5e-10	1e9	-0.5
gidney_cultivation	teleport_d3_p0.001_s	acceptance	0.646 ± 2e-05	0.646 ± 2e-06	1e9	-1.7
gidney_cultivation	teleport_d3_p0.001_s	LER	5.42e-07 ± 3e-08	5.89e-07 ± 4e-09	1e9	+1.6
gidney_cultivation	teleport_d3_p0.002_s	acceptance	0.418 ± 2e-05	0.418 ± 5e-06	1e9	-0.6
gidney_cultivation	teleport_d3_p0.002_s	LER	4.86e-06 ± 1e-07	4.83e-06 ± 3e-08	1e9	-0.3
gidney_cultivation	unitary_d3_p0.0005_s	acceptance	0.829 ± 1e-05	0.829 ± 5e-07	1e9	-2.0
gidney_cultivation	unitary_d3_p0.0005_s	LER	3.14e-08 ± 6e-09	4.11e-08 ± 3e-10	1e9	+1.6
gidney_cultivation	unitary_d3_p0.001_s	acceptance	0.687 ± 1e-05	0.687 ± 2e-06	1e9	-0.3
gidney_cultivation	unitary_d3_p0.001_s	LER	3.54e-07 ± 2e-08	3.43e-07 ± 2e-09	1e9	-0.5
gidney_cultivation	unitary_d3_p0.001_z	acceptance	0.687 ± 1e-05	0.687 ± 1e-06	1e9	+1.0
gidney_cultivation	unitary_d3_p0.001_z	LER	2.87e-07 ± 2e-08	2.88e-07 ± 2e-09	1e9	+0.1
gidney_cultivation	unitary_d3_p0.002_s	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1e9	+1.1
gidney_cultivation	unitary_d3_p0.002_s	LER	2.98e-06 ± 8e-08	2.95e-06 ± 2e-08	1e9	-0.3
gidney_cultivation	unitary_d3_p0.002_z	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1e9	-0.5
gidney_cultivation	unitary_d3_p0.002_z	LER	2.53e-06 ± 7e-08	2.46e-06 ± 2e-08	1e9	-0.8
gidney_cultivation	unitary_d3_p0.003_s	acceptance	0.325 ± 1e-05	0.325 ± 6e-06	1e9	+0.1
gidney_cultivation	unitary_d3_p0.003_s	LER	1.04e-05 ± 2e-07	1.05e-05 ± 7e-08	1e9	+0.5
gidney_cultivation	unitary_d3_p0.003_z	acceptance	0.325 ± 1e-05	0.325 ± 6e-06	1e9	+0.9
gidney_cultivation	unitary_d3_p0.003_z	LER	8.92e-06 ± 2e-07	9e-06 ± 6e-08	1e9	+0.5
gidney_cultivation	unitary_d3_p0.005_s	acceptance	0.155 ± 1e-05	0.155 ± 6e-06	1e9	+0.3
gidney_cultivation	unitary_d3_p0.005_s	LER	5.65e-05 ± 6e-07	5.69e-05 ± 3e-07	1e9	+0.6
gidney_cultivation	unitary_d3_p0.005_z	acceptance	0.155 ± 1e-05	0.155 ± 6e-06	1e9	-0.0

study	point	quantity	Stim (CPU)	Tsim GPU	Stim shots	z
gidney_cultivation	unitary_d3_p0.005_z	LER	4.84e-05 ± 6e-07	4.84e-05 ± 3e-07	1e9	-0.0
gidney_cultivation	unitary_d3_p0.007_s	acceptance	0.0742 ± 8e-06	0.0742 ± 3e-06	1e9	+1.0
gidney_cultivation	unitary_d3_p0.007_s	LER	0.000183 ± 2e-06	0.000184 ± 7e-07	1e9	+0.8
gidney_cultivation	unitary_d3_p0.007_z	acceptance	0.0743 ± 8e-06	0.0743 ± 3e-06	1e9	+1.7
gidney_cultivation	unitary_d3_p0.007_z	LER	0.000156 ± 1e-06	0.000154 ± 6e-07	1e9	-1.0
gidney_cultivation	unitary_d3_p0.01_s	acceptance	0.0249 ± 5e-06	0.0249 ± 4e-06	1e9	+0.5
gidney_cultivation	unitary_d3_p0.01_s	LER	0.000692 ± 5e-06	0.000685 ± 4e-06	1e9	-1.0
gidney_cultivation	unitary_d3_p0.01_z	acceptance	0.025 ± 5e-06	0.025 ± 4e-06	1e9	-0.2
gidney_cultivation	unitary_d3_p0.01_z	LER	0.000579 ± 5e-06	0.000585 ± 4e-06	1e9	+0.8
wan_few_clifford_terms	d3_p0.0005_s	acceptance	0.804 ± 1e-05	0.804 ± 7e-07	1e9	+0.0
wan_few_clifford_terms	d3_p0.0005_s	LER	5.97e-08 ± 9e-09	7.18e-08 ± 5e-10	1e9	+1.4
wan_few_clifford_terms	d3_p0.0015_s	acceptance	0.519 ± 2e-05	0.519 ± 4e-06	1e9	+1.3
wan_few_clifford_terms	d3_p0.0015_s	LER	2.01e-06 ± 6e-08	2.02e-06 ± 1e-08	1e9	+0.1
wan_few_clifford_terms	d3_p0.001_s	acceptance	0.646 ± 2e-05	0.646 ± 2e-06	1e9	-0.7
wan_few_clifford_terms	d3_p0.001_s	LER	5.65e-07 ± 3e-08	5.81e-07 ± 4e-09	1e9	+0.5
wan_few_clifford_terms	d3_p0.0025_s	acceptance	0.336 ± 1e-05	0.336 ± 6e-06	1e9	+0.6
wan_few_clifford_terms	d3_p0.0025_s	LER	9.64e-06 ± 2e-07	9.67e-06 ± 7e-08	1e9	+0.1
wan_few_clifford_terms	d3_p0.002_s	acceptance	0.418 ± 2e-05	0.418 ± 5e-06	1e9	+1.4
wan_few_clifford_terms	d3_p0.002_s	LER	4.87e-06 ± 1e-07	4.85e-06 ± 3e-08	1e9	-0.2
wan_few_clifford_terms	d3_p0.003_s	acceptance	0.27 ± 1e-05	0.27 ± 6e-06	1e9	+0.3
wan_few_clifford_terms	d3_p0.003_s	LER	1.68e-05 ± 2e-07	1.69e-05 ± 1e-07	1e9	+0.4
wan_few_clifford_terms	d3_p0.004_s	acceptance	0.175 ± 1e-05	0.175 ± 6e-06	1e9	-0.6
wan_few_clifford_terms	d3_p0.004_s	LER	4.2e-05 ± 5e-07	4.21e-05 ± 2e-07	1e9	+0.3
wan_few_clifford_terms	d3_p0.005_s	acceptance	0.114 ± 1e-05	0.114 ± 6e-06	1e9	-0.6
wan_few_clifford_terms	d3_p0.005_s	LER	8.56e-05 ± 9e-07	8.47e-05 ± 5e-07	1e9	-0.9
gauging_cultivation	d3_p0.0005_s	acceptance	0.679 ± 1e-05	0.679 ± 9e-07	1e9	-0.1
gauging_cultivation	d3_p0.0005_s	LER	9.73e-08 ± 1e-08	1.07e-07 ± 8e-10	1e9	+0.8
gauging_cultivation	d3_p0.001_s	acceptance	0.461 ± 2e-05	0.461 ± 2e-06	1e9	-0.9
gauging_cultivation	d3_p0.001_s	LER	9.12e-07 ± 4e-08	8.77e-07 ± 6e-09	1e9	-0.8
gauging_cultivation	d3_p0.002_s	acceptance	0.213 ± 1e-05	0.213 ± 4e-06	1e9	+0.2
gauging_cultivation	d3_p0.002_s	LER	7.7e-06 ± 2e-07	7.58e-06 ± 5e-08	1e9	-0.6
gauging_cultivation	d3_p0.003_s	acceptance	0.0983 ± 9e-06	0.0983 ± 3e-06	1e9	+1.1
gauging_cultivation	d3_p0.003_s	LER	2.65e-05 ± 5e-07	2.76e-05 ± 2e-07	1e9	+1.9
gauging_cultivation	d3_p0.004_s	acceptance	0.0456 ± 7e-06	0.0456 ± 2e-06	1e9	+0.7
gauging_cultivation	d3_p0.004_s	LER	6.92e-05 ± 1e-06	7.04e-05 ± 4e-07	1e9	+1.0
gauging_cultivation	d3_p0.005_s	acceptance	0.0211 ± 5e-06	0.0212 ± 1e-06	1e9	+1.1
gauging_cultivation	d3_p0.005_s	LER	0.000147 ± 3e-06	0.000148 ± 7e-07	1e9	+0.3
fold_transversal	fd3_p0.0005_y	acceptance	0.897 ± 1e-05	0.897 ± 3e-07	1e9	-0.4
fold_transversal	fd3_p0.0005_y	LER	7.81e-09 ± 3e-09	6.34e-09 ± 8e-11	1e9	-0.5
fold_transversal	fd3_p0.0007_y	acceptance	0.858 ± 1e-05	0.858 ± 3e-07	1e9	+0.1
fold_transversal	fd3_p0.0007_y	LER	2.45e-08 ± 5e-09	1.74e-08 ± 1e-10	1e9	-1.3
fold_transversal	fd3_p0.0015_y	acceptance	0.721 ± 1e-05	0.721 ± 1e-06	1e9	+1.7
fold_transversal	fd3_p0.0015_y	LER	1.44e-07 ± 1e-08	1.74e-07 ± 1e-09	1e9	+2.1
fold_transversal	fd3_p0.001_y	acceptance	0.804 ± 1e-05	0.804 ± 6e-07	1e9	-0.6
fold_transversal	fd3_p0.001_y	LER	4.98e-08 ± 8e-09	5.11e-08 ± 4e-10	1e9	+0.2
fold_transversal	fd3_p0.002_y	acceptance	0.647 ± 2e-05	0.647 ± 2e-06	1e9	-0.7
fold_transversal	fd3_p0.002_y	LER	3.62e-07 ± 2e-08	4.11e-07 ± 3e-09	1e9	+2.1
fold_transversal	fd3_p0.003_y	acceptance	0.52 ± 2e-05	0.52 ± 3e-06	1e9	-0.1
fold_transversal	fd3_p0.003_y	LER	1.46e-06 ± 5e-08	1.42e-06 ± 1e-08	1e9	-0.7
chan_flags	d3a6_p0.0005_s	acceptance	0.829 ± 1e-05	0.829 ± 8e-07	1e9	-0.2
chan_flags	d3a6_p0.0005_s	LER	4.1e-08 ± 7e-09	4.13e-08 ± 5e-10	1e9	+0.0
chan_flags	d3a6_p0.001_s	acceptance	0.687 ± 1e-05	0.687 ± 2e-06	1e9	-0.0
chan_flags	d3a6_p0.001_s	LER	3.17e-07 ± 2e-08	3.42e-07 ± 2e-09	1e9	+1.1
chan_flags	d3a6_p0.002_s	acceptance	0.472 ± 2e-05	0.472 ± 4e-06	1e9	+1.1
chan_flags	d3a6_p0.002_s	LER	2.86e-06 ± 8e-08	2.93e-06 ± 2e-08	1e9	+0.9
chan_flags	d3a6_p0.003_s	acceptance	0.325 ± 1e-05	0.325 ± 6e-06	1e9	+2.0
chan_flags	d3a6_p0.003_s	LER	1.07e-05 ± 2e-07	1.07e-05 ± 8e-08	1e9	+0.4
chan_flags	d3a6_p0.005_s	acceptance	0.155 ± 1e-05	0.155 ± 6e-06	1e9	+2.1
chan_flags	d3a6_p0.005_s	LER	5.73e-05 ± 6e-07	5.74e-05 ± 3e-07	1e9	+0.1
chan_flags	d3a6_p0.007_s	acceptance	0.0742 ± 8e-06	0.0742 ± 3e-06	1e9	-0.4

study	point	quantity	Stim (CPU)	Tsim GPU	Stim shots	z
chan_flags	d3a6_p0.007_s	LER	0.000186 ± 2e-06	0.000183 ± 7e-07	1e9	-2.0
chan_flags	d3a6_p0.01_s	acceptance	0.0249 ± 5e-06	0.0249 ± 4e-06	1e9	-0.6
chan_flags	d3a6_p0.01_s	LER	0.000685 ± 5e-06	0.000682 ± 4e-06	1e9	-0.5
chan_flags	d3a6f2_p0.0005_s	acceptance	0.799 ± 1e-05	0.799 ± 8e-07	1e9	+0.5
chan_flags	d3a6f2_p0.0005_s	LER	5.25e-08 ± 8e-09	4.74e-08 ± 5e-10	1e9	-0.6
chan_flags	d3a6f2_p0.001_s	acceptance	0.639 ± 2e-05	0.639 ± 2e-06	1e9	+0.5
chan_flags	d3a6f2_p0.001_s	LER	3.63e-07 ± 2e-08	3.99e-07 ± 3e-09	1e9	+1.5
chan_flags	d3a6f2_p0.002_s	acceptance	0.409 ± 2e-05	0.409 ± 4e-06	1e9	-0.5
chan_flags	d3a6f2_p0.002_s	LER	3.48e-06 ± 9e-08	3.5e-06 ± 2e-08	1e9	+0.2
chan_flags	d3a6f2_p0.003_s	acceptance	0.263 ± 1e-05	0.263 ± 5e-06	1e9	-0.3
chan_flags	d3a6f2_p0.003_s	LER	1.27e-05 ± 2e-07	1.28e-05 ± 8e-08	1e9	+0.5
chan_flags	d3a6f2_p0.005_s	acceptance	0.109 ± 1e-05	0.109 ± 5e-06	1e9	+1.7
chan_flags	d3a6f2_p0.005_s	LER	7.21e-05 ± 8e-07	7.16e-05 ± 4e-07	1e9	-0.5
chan_flags	d3a6f2_p0.007_s	acceptance	0.0454 ± 7e-06	0.0453 ± 3e-06	1e9	-1.0
chan_flags	d3a6f2_p0.007_s	LER	0.00024 ± 2e-06	0.000239 ± 1e-06	1e9	-0.4
chan_flags	d3a6f2_p0.01_s	acceptance	0.0124 ± 4e-06	0.0124 ± 1e-06	1e9	-0.2
chan_flags	d3a6f2_p0.01_s	LER	0.000938 ± 9e-06	0.000934 ± 4e-06	1e9	-0.5
magna	bh10-S_p0.0001	acceptance	0.996 ± 2e-06	0.996 ± 5e-07	1e9	-1.0
magna	bh10-S_p0.0001	LER	1.33e-06 ± 4e-08	1.4e-06 ± 1e-08	1e9	+1.9
magna	bh10-S_p0.0002	acceptance	0.992 ± 3e-06	0.992 ± 1e-06	1e9	+0.4
magna	bh10-S_p0.0002	LER	5.56e-06 ± 7e-08	5.49e-06 ± 4e-08	1e9	-0.9
magna	bh10-S_p0.0005	acceptance	0.981 ± 4e-06	0.981 ± 5e-06	1e9	-1.2
magna	bh10-S_p0.0005	LER	3.51e-05 ± 2e-07	3.48e-05 ± 2e-07	1e9	-1.1
magna	bh10-S_p0.001	acceptance	0.963 ± 6e-06	0.963 ± 1e-05	1e9	-0.3
magna	bh10-S_p0.001	LER	0.000141 ± 4e-07	0.00014 ± 6e-07	1e9	-0.9
magna	bh2-S_p0.0001	acceptance	0.999 ± 1e-06	0.999 ± 7e-08	1e9	+0.0
magna	bh2-S_p0.0001	LER	5.81e-08 ± 8e-09	6.92e-08 ± 5e-10	1e9	+1.5
magna	bh2-S_p0.0002	acceptance	0.997 ± 2e-06	0.997 ± 2e-07	1e9	+0.7
magna	bh2-S_p0.0002	LER	2.52e-07 ± 2e-08	2.77e-07 ± 2e-09	1e9	+1.6
magna	bh2-S_p0.0005	acceptance	0.993 ± 3e-06	0.993 ± 8e-07	1e9	+1.4
magna	bh2-S_p0.0005	LER	1.81e-06 ± 4e-08	1.75e-06 ± 1e-08	1e9	-1.5
magna	bh2-S_p0.001	acceptance	0.986 ± 4e-06	0.986 ± 2e-06	1e9	-1.4
magna	bh2-S_p0.001	LER	7.1e-06 ± 8e-08	7.06e-06 ± 4e-08	1e9	-0.4
magna	bh6-S_p0.0001	acceptance	0.997 ± 2e-06	0.997 ± 3e-07	1e9	+0.3
magna	bh6-S_p0.0001	LER	4.76e-07 ± 2e-08	4.91e-07 ± 3e-09	1e9	+0.7
magna	bh6-S_p0.0002	acceptance	0.995 ± 2e-06	0.995 ± 7e-07	1e9	-2.0
magna	bh6-S_p0.0002	LER	2.04e-06 ± 5e-08	1.94e-06 ± 1e-08	1e9	-2.1
magna	bh6-S_p0.0005	acceptance	0.987 ± 4e-06	0.987 ± 3e-06	1e9	-0.1
magna	bh6-S_p0.0005	LER	1.21e-05 ± 1e-07	1.23e-05 ± 8e-08	1e9	+1.5
magna	bh6-S_p0.001	acceptance	0.974 ± 5e-06	0.974 ± 5e-06	1e9	+0.6
magna	bh6-S_p0.001	LER	4.95e-05 ± 2e-07	4.91e-05 ± 2e-07	1e9	-1.5
magna	rp2-d3-S_p0.001	acceptance	0.514 ± 2e-05	0.514 ± 2e-06	1e9	+2.5
magna	rp2-d3-S_p0.001	LER	8.6e-07 ± 4e-08	8.34e-07 ± 6e-09	1e9	-0.6
magna	rp2-d5-S_p0.001	acceptance	0.0945 ± 9e-06	0.0945 ± 2e-07	1e9	+0.2
rp2_ungrown	msc3_s	acceptance	0.514 ± 2e-05	0.514 ± 2e-06	1e9	+1.8
rp2_ungrown	msc3_s	LER	7.59e-07 ± 4e-08	8.25e-07 ± 6e-09	1e9	+1.7
erasure_injection	hook_p0.0001_s	acceptance	0.99 ± 3e-06	0.99 ± 3e-06	1e9	+0.1
erasure_injection	hook_p0.0001_s	LER	5.34e-05 ± 2e-07	5.31e-05 ± 2e-07	1e9	-0.9
erasure_injection	hook_p0.001_s	acceptance	0.902 ± 9e-06	0.902 ± 2e-05	1e9	-0.9
erasure_injection	hook_p0.001_s	LER	0.000547 ± 8e-07	0.000546 ± 1e-06	1e9	-0.7

1 Gidney, Shutty, Jones (2024), Magic state cultivation: growing T states as cheap as CNOT gates

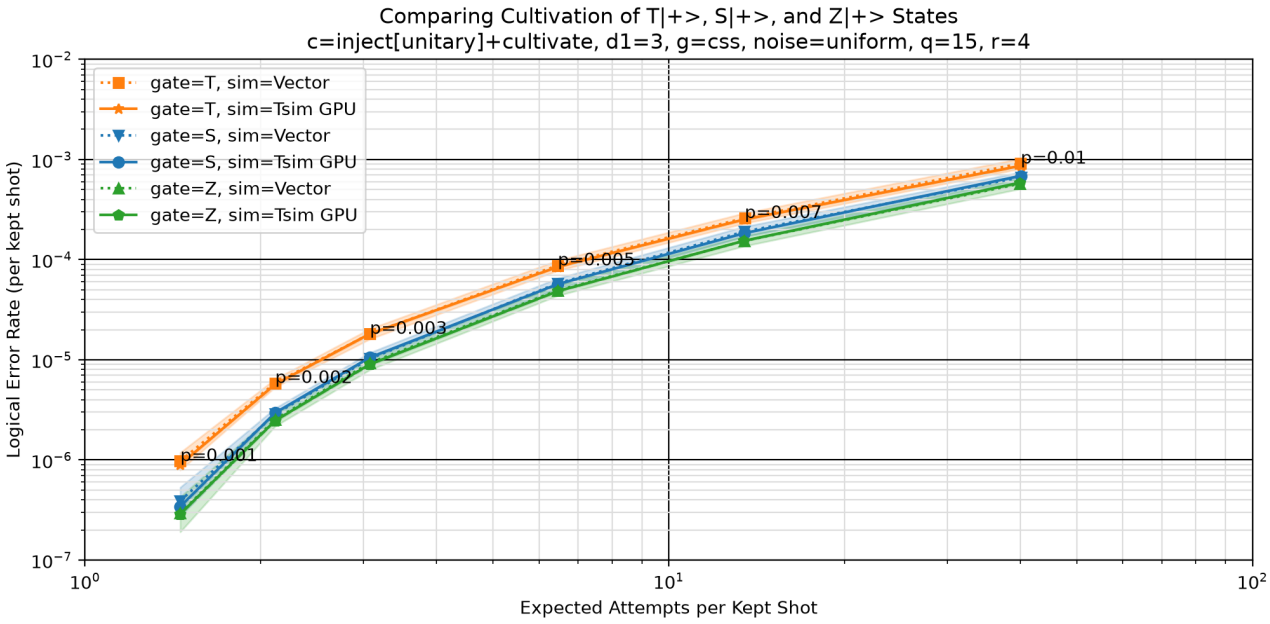
arXiv:2409.17595 Study module `repro/studies/gidney_cultivation.py`

Two figures, both decoder-free (all detectors postselected, noiseless final logical check):

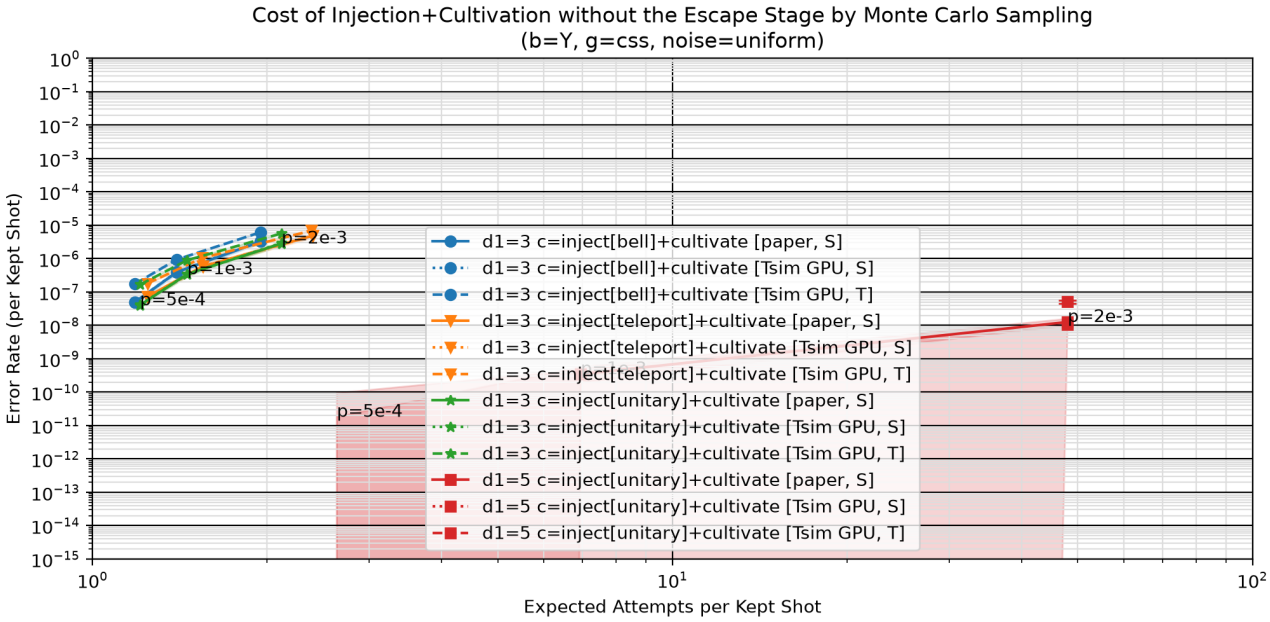
- Fig. 13 (`t-vs-s`): unitary injection + d=3 cultivation with each T gate replaced by T, S or Z, vs. uniform depolarizing noise p. The paper used a state-vector simulator ("vec_intercept").
- Fig. 21 (`inject-only-sampled`): bell / teleport ("degenerate") / unitary injection + d=3 cultivation, and unitary + d=5 cultivation, at p = 5e-4, 1e-3, 2e-3. The paper sampled the S proxy with Stim ("perfectionist" decoder = discard on any detection event).

Circuits: the authors' generator (`cultiv.make_inject_and_cultivate_circuit`, commit 871e68f) with their `gen.NoiseModel.uniform_depolarizing(p)`; T circuits by the Clifft paper's converter (`convert_circuit`: S->T, d=5 sign flips and feedforward, final Y_L wrapped in T...T_DAG). Z variant: S/S_DAG -> Z and the final logical check measures X_L instead of Y_L.

Published numbers overlaid: Zenodo 13777072 `stats.csv`.



Gidney et al. Fig. 13: cultivating $T|+\rangle$, $S|+\rangle$, $Z|+\rangle$ states ($d = 3$, no escape)
Compute: 1.6 min GPU sampling + 5.3 s compile, 3.1e11 shots, 18 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Drawn with the authors' own `sinter plot` command (`step3_plot`) on the paper's released stats (sim=Vector: their state-vector simulation) plus ours (sim=Tsim GPU). Every detector postselected; noiseless final logical check. S and Z are Clifford proxies of the same circuit.



Gidney et al. Fig. 21: injection + cultivation without escape, Monte Carlo sampling
Compute: 28.0 min GPU sampling + 12 s compile, 7.3e12 shots, 21 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Drawn with the authors' own `sinter plot` command on the paper's released stats ([paper, S]: Stim, S proxy, up to 10^{12} shots per point) plus ours: [Tsim GPU, S] the same S proxy on the GPU, [Tsim GPU, T] the real T circuit. Shaded: hypotheses within a factor 1000 of the maximum likelihood.

Gidney et al.: $d = 5$ injection + cultivation (no escape)
Each cell: logical error rate of kept shots, errors / kept shots, discard rate, and (Tsim) GPU sampling time.
Compute: 14.0 min GPU sampling + 6.3 s compile, 5.1e12 shots, 3 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

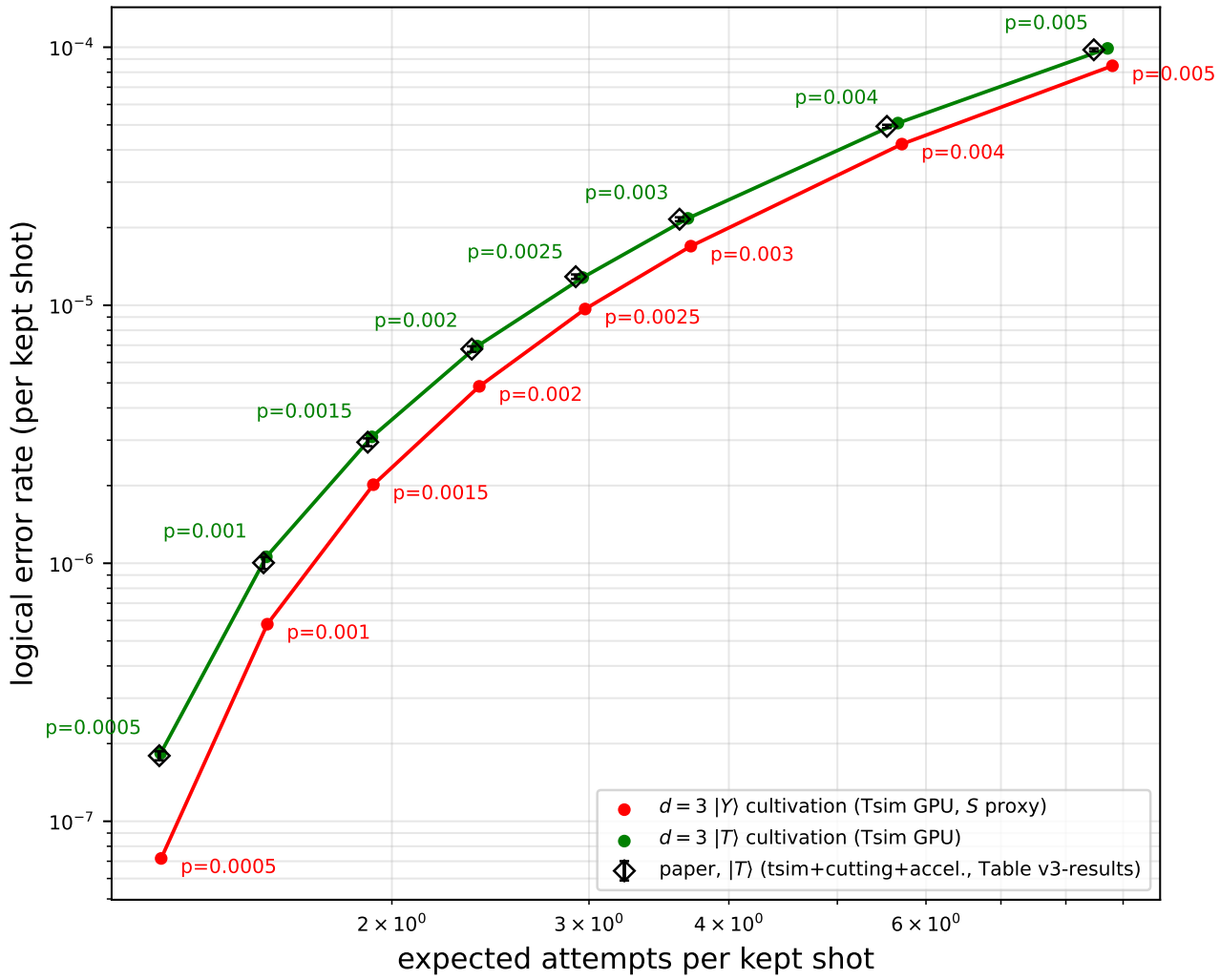
p	paper (S proxy, Stim)	Tsim GPU, S proxy	Tsim GPU, T
0.0005	0 (0/3.8e11, disc. 0.6208)	0 (0/7.4e10, disc. 0.6208), 1.0 min	–
0.001	3.47e-10 (50/1.4e11, disc. 0.8560)	–	–
0.002	1.26e-08 (263/2.1e10, disc. 0.9792)	1.05e-08 (874/8.3e10, disc. 0.9792), 10.7 min	5.07e-08 (1000/2e10, disc. 0.9792), 2.3 min

2 Wan, Zhong, Zapirain (2025), Simulating magic state cultivation with few Clifford terms

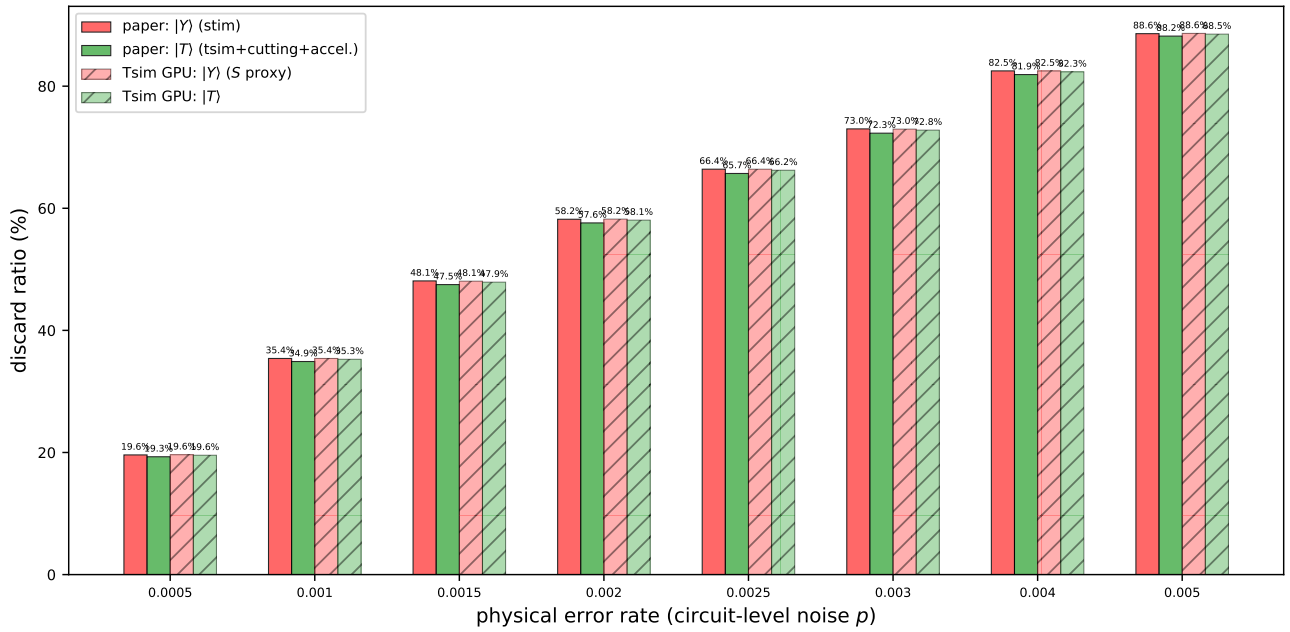
arXiv:2509.08658 Study module `repro/studies/wan_few_clifford_terms.py`

The paper samples the native-T $d = 3$ cultivation circuit (injection + double check, from the companion repo's `d_3_circuit_definitions.py`) followed by a noiseless T-projection readout, postselecting on every detector. Its Table (`tab:v3-results`) lists the exact non-Clifford results at 2^{29}–2^{32} shots per noise strength; its red curves are the S-gate proxy sampled with Stim.

Circuits: built as in the repo's `run.py` and our `make_circuits.py`: $T \to S$, the vendored `gen.NoiseModel.uniform_depolar` $S \to T$; the projection stays noiseless. The Clifford proxy maps every $T/T^\dagger$ of the whole circuit (including the projection) to $S/S^\dagger$.



Wan et al. Figs. 15 and 17: $d = 3$ T cultivation, LER vs. expected attempts per kept shot
Compute: 4.5 min GPU sampling + 5.3 s compile, 6.5e11 shots, 16 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same format as the paper (shaded bands: binomial standard error). Red: the $|Y\rangle$ (S-gate) proxy, which the paper sampled with Stim; green: real T gates; open diamonds: the paper's exact non-Clifford results (Table v3-results, 2^{29}–2^{32} shots, Apple M4 Pro CPU). The T LERs agree with the table at every p.



Wan et al. Fig. 19: discard ratio (1 − PSR)
Compute: 4.5 min GPU sampling + 5.3 s compile, 6.5e11 shots, 16 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Solid bars: the values printed on the paper's figure; hatched: Tsim GPU on the circuits built from the authors' repo. The S proxy reproduces the paper's Stim bars exactly. For the T circuit the paper discards 0.3–0.6 points less than for its S proxy, while Tsim GPU discards only about 0.1 point less, so the two T samplers (the paper's Tsim 0.1.0 with cutting, and the CUDA kernel here) disagree on the T post-selection rate by a few 10^{-3}, far beyond statistics, while their LERs agree. A third, independent exact simulator sides with the GPU: Clifft 0.13 on the same T circuit at $p = 10^{-3}$ keeps 0.64726 ± 0.00008 of 4×10^7 shots (Tsim GPU 0.64732; paper 0.6511), and 0.64595 for the S proxy (Tsim GPU 0.64600).

Wan et al. Table v3-results (left, as published) vs. Tsim GPU (right)

Compute: 1.3 min GPU sampling + 3.0 s compile, 2e11 shots, 8 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

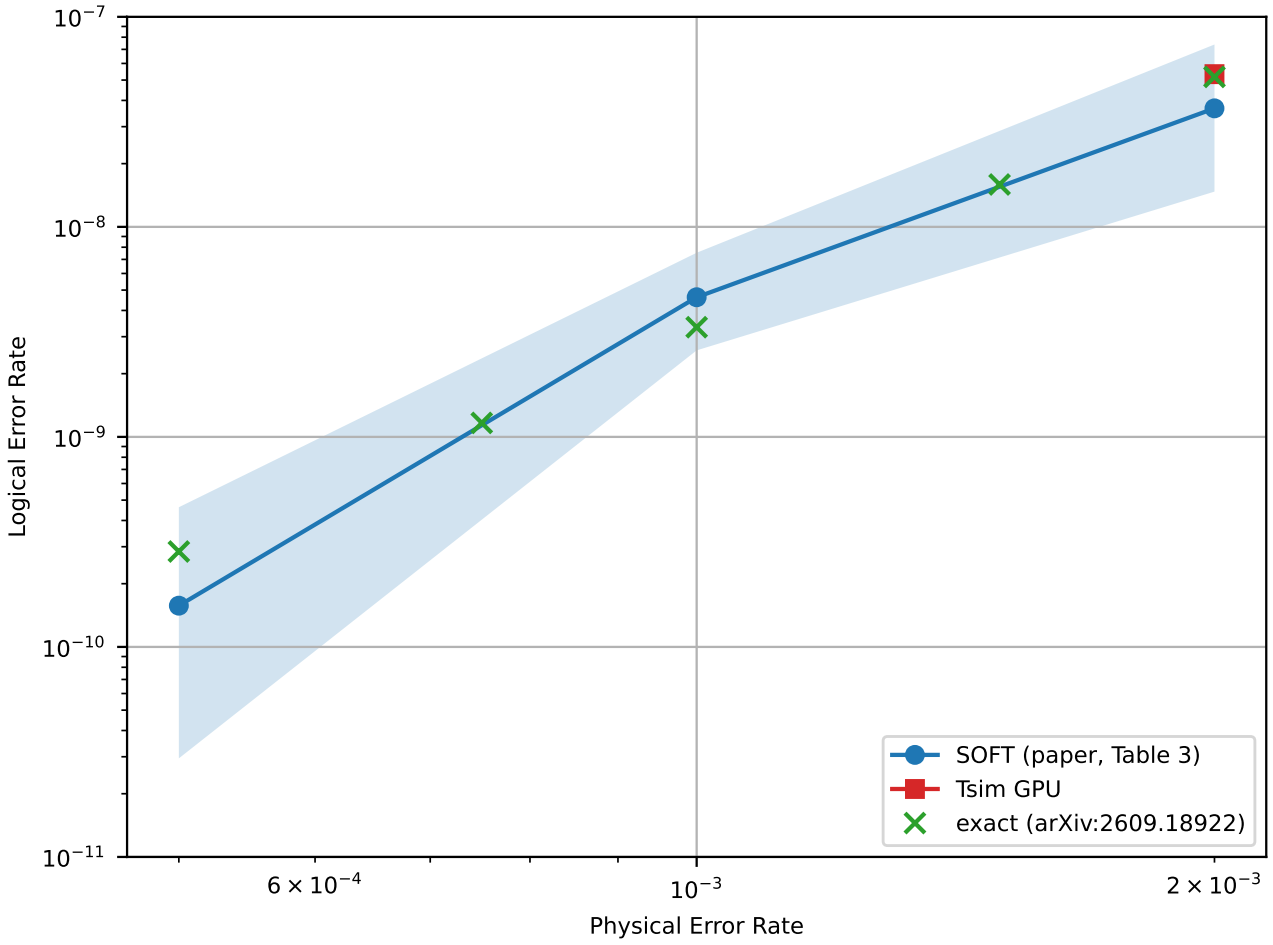
paper (Table v3-results, T, Apple M4 Pro CPU)						Tsim GPU (T)				
p	**Shots**	**PSR**	**Errors**	n_{kept}	**LER**	LER	PSR	errors	shots	GPU time
5×10^{-4}	2^{32}	0.807	622	3,465,069,335	1.8e-07	1.83e-07	0.8045	20029	1.4e11	60 s
1×10^{-3}	2^{29}	0.651	351	349,529,061	1e-06	1.06e-06	0.6473	20034	2.9e10	11 s
1.5×10^{-3}	2^{29}	0.525	831	282,110,746	2.9e-06	3.09e-06	0.5210	20479	1.3e10	4.1 s
2×10^{-3}	2^{29}	0.424	1,540	227,748,263	6.8e-06	6.94e-06	0.4194	20425	7e9	2.1 s
2.5×10^{-3}	2^{29}	0.343	2,373	183,908,903	1.3e-05	1.28e-05	0.3377	20794	4.8e9	1.3 s
3×10^{-3}	2^{29}	0.277	3,201	148,548,040	2.2e-05	2.17e-05	0.2720	20768	3.5e9	1.0 s
4×10^{-3}	2^{29}	0.181	4,785	96,970,177	4.9e-05	5.08e-05	0.1766	23758	2.6e9	0.8 s
5×10^{-3}	2^{29}	0.118	6,193	63,357,051	9.8e-05	9.9e-05	0.1147	37503	3.3e9	0.8 s

3 Li et al. (2025), SOFT: a high-performance simulator for universal fault-tolerant quantum circuits

arXiv:2512.23037 Study module `repro/studies/soft_d5.py`

SOFT sampled the authors' own native-T circuits (unitary injection, $d = 3$ and $d = 5$ double checks, code growth, noiseless readout; every detector postselected) on 16 H800 GPUs for 20 days: 28.9B, 74B and 134.4B shots at p = 2e-3, 1e-3, 5e-4. SOFT ran no Clifford proxy, so only the non-Clifford version is sampled here (the S proxy of the same protocol is in the Gidney study).

Circuits: SOFT's released files (`softv1_magic_state_cultivation`), used verbatim. They are also the reference circuits of the exact-LER paper (Wan, Zapirain, arXiv:2609.18922), whose exact values $A(p)$ and $P_L(p)$ and truncated series are overlaid.



SOFT Fig. 2: logical error rate of $d = 5$ magic state injection + cultivation (real T gates)
Compute: 7.7 min GPU sampling + 4.2 s compile, 2.8e12 shots, 4 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same axes as the paper. Blue: SOFT's published points (16 H800 GPUs, 20 days, 2.4×10^{11} shots); red: Tsim GPU on SOFT's own circuits; green crosses: exact values of Wan and Zapirain (arXiv:2609.18922) for the same circuits. Bands: hypotheses with a Bayes factor of at most 1000 relative to the maximum-likelihood hypothesis (binomial), as in the paper. Our points at $p > 2 \times 10^{-3}$ are in the table below.

SOFT Table 3: performance of $d = 5$ cultivation (1 B = 1 billion), paper vs. Tsim GPU vs. exact
Same columns as the paper's Table 3, with our rows (and the exact values) added.
Compute: 7.9 min GPU sampling + 4.5 s compile, 2.8e12 shots, 5 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

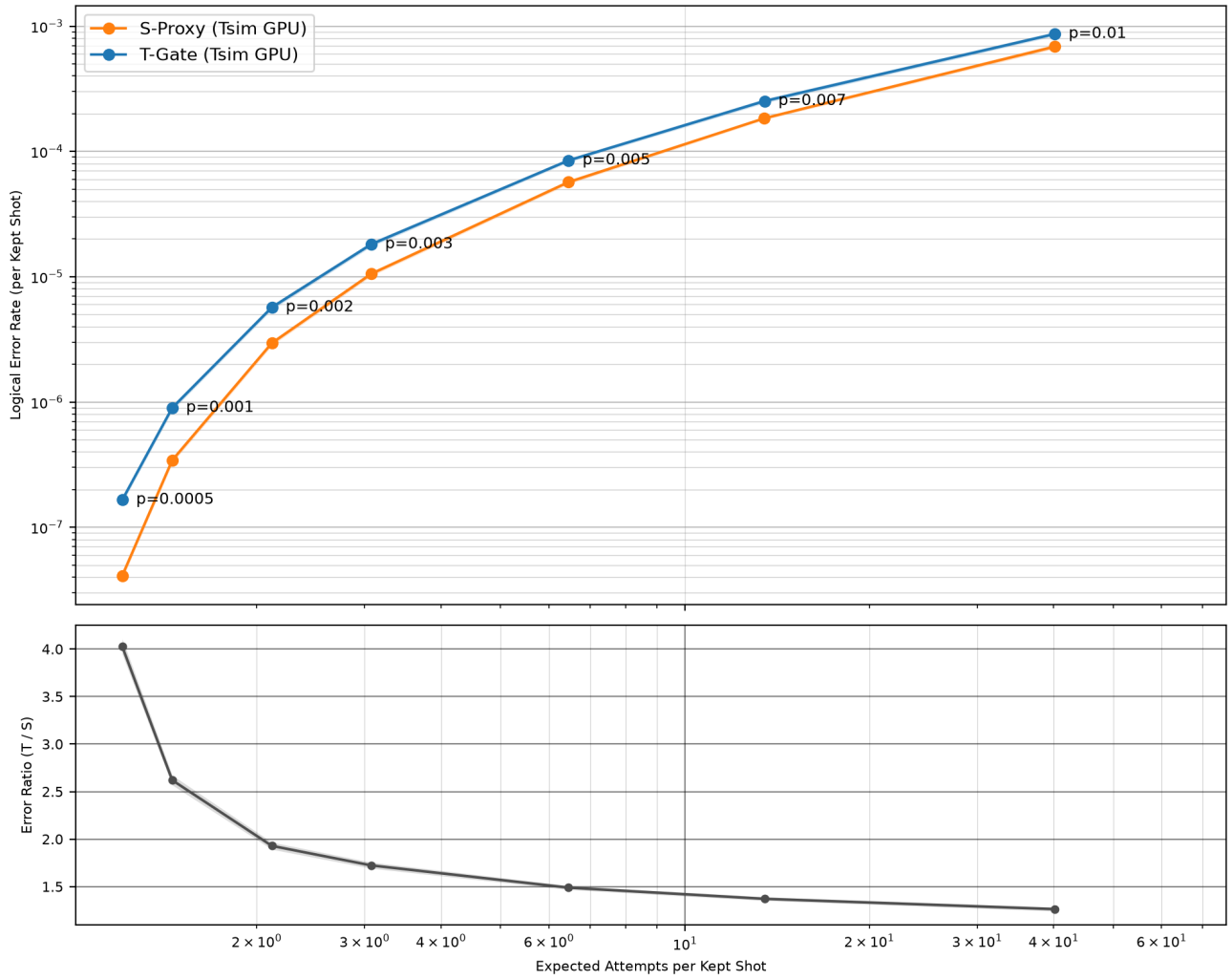
Noise Strength		Total Shots	Preserved Shots	Discard Rate	Detected Logical Errors	Logical Error Rate	hardware, time
0.005	Tsim GPU	600.0B	0.04B	99.99%	116	2.97×10^{-6}	1 RTX 5090, 2.1 min
0.004	Tsim GPU	600.0B	0.27B	99.96%	295	1.11×10^{-6}	1 RTX 5090, 1.7 min
0.003	Tsim GPU	600.0B	1.82B	99.70%	555	3.05×10^{-7}	1 RTX 5090, 1.4 min
0.002	SOFT (paper)	28.9B	0.60B	97.92%	22	3.67×10^{-8}	16 H800, ∼20 days total
0.002	Tsim GPU	1000.0B	20.83B	97.92%	1111	5.33×10^{-8}	1 RTX 5090, 2.5 min
0.002	exact (2609.18922)			97.92%		5.18×10^{-8}	
0.001	SOFT (paper)	74.0B	10.60B	85.68%	49	4.62×10^{-9}	16 H800, ∼20 days total
0.001	exact (2609.18922)			85.60%		3.33×10^{-9}	
0.0005	SOFT (paper)	134.4B	50.90B	62.13%	8	1.57×10^{-10}	16 H800, ∼20 days total
0.0005	exact (2609.18922)			62.08%		2.85×10^{-10}	

4 Chase, Labib (2026), Clifft: fast exact simulation of near-Clifford quantum circuits

arXiv:2604.27058 Study module `repro/studies/clifft_inject_cultivate.py`

Clifft's Fig. 6 plots the logical error rate per kept shot of Gidney's unitary-injection cultivation circuits (no escape) with real T gates and with the S proxy against the expected attempts per kept shot, with the T/S ratio below; its Table 2 lists the $d = 5$ T-gate values at three noise strengths (stratified importance sampling, 9.6×10^{10} shots, 12 machine-hours on a 16-core c6i.8xlarge).

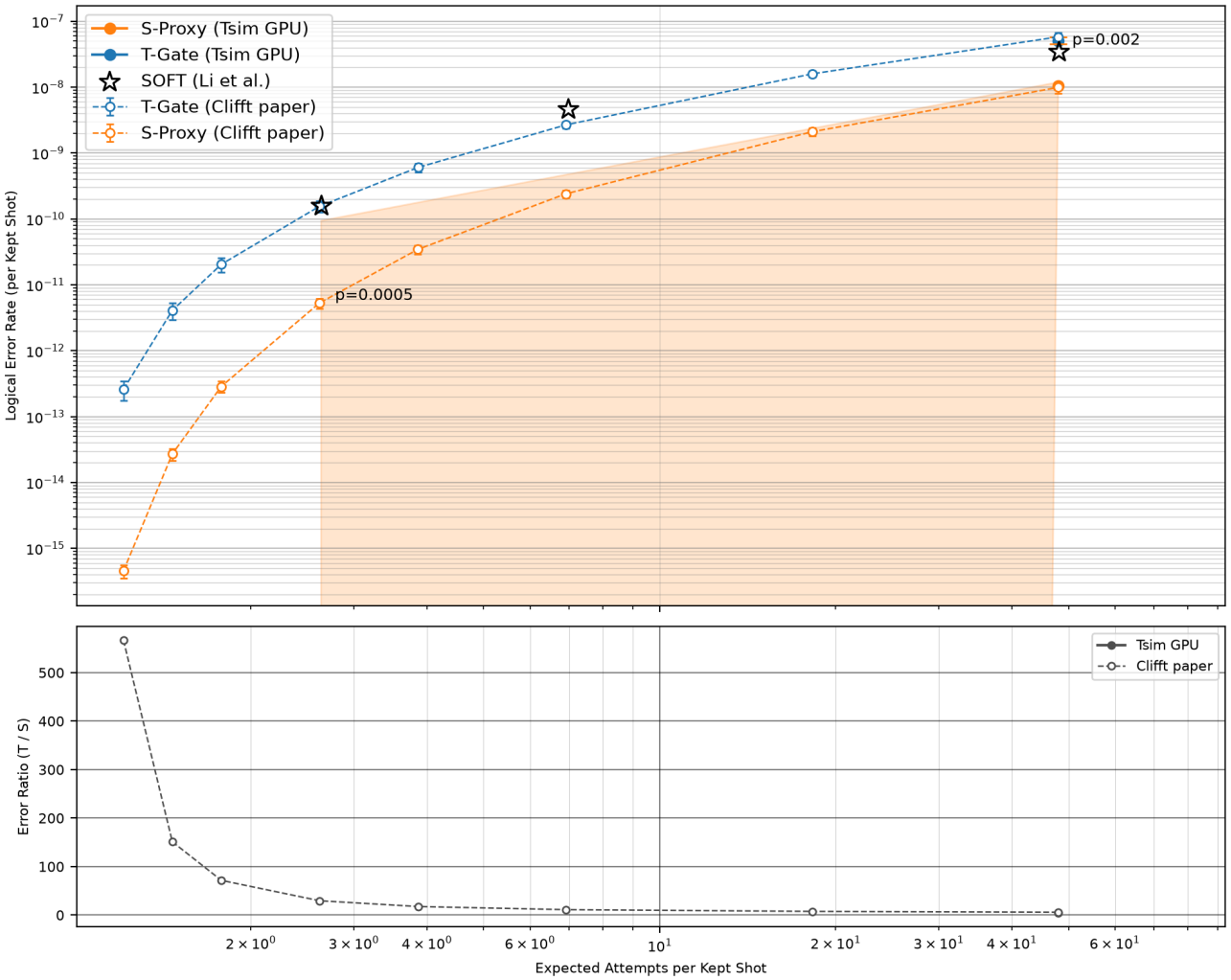
Clifft's circuits are the Clifft paper's converter applied to Gidney's generator, the same as the Gidney study's (byte-identical at $p = 10^{-3}$), so this study samples nothing itself: it re-plots the Gidney study's unitary-injection points. The figures are drawn with the plotting code of the Clifft paper (unitaryfoundation/clifft-paper at db7dc9f, `magic_state_cultivation/plot_results.py`: `_plot_ic_single` and `_compute_ic_ratio`, Apache-2.0), fed with our points instead of their importance-sampling strata; their published values (`results/inject_cultivate/analysis.jso` are overlaid.



Clifft Fig. 6(a): inject + cultivate, $d = 3$ colour code, T vs. S proxy

Compute: (Gidney study points): 5.6 min GPU sampling + 4.0 s compile, 9e11 shots, 14 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

Drawn with the Clifft paper's own plotting code (`plot_results.py`) on the Gidney study's samples (filled, solid; shaded: 1000x likelihood). Open markers, dashed: the Clifft paper's published importance-sampling values (`analysis.json`; $d = 3$ values are not in it). Bottom: T/S ratio, median and 95% credible interval (bootstrap, as in the paper).



Clifft Fig. 6(b): inject + cultivate, $d = 5$ colour code, T vs. S proxy
Compute: (Gidney study points): 14.0 min GPU sampling + 6.3 s compile, 5.1e12 shots, 3 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Drawn with the Clifft paper's own plotting code (`plot_results.py`) on the Gidney study's samples (filled, solid; shaded: 1000x likelihood). Open markers, dashed: the Clifft paper's published importance-sampling values (`analysis.json`, $d = 5$ only). Stars: SOFT. Bottom: T/S ratio, median and 95% credible interval (bootstrap, as in the paper).

5 Hetényi, Brown, Williamson (2026), Constant-depth magic state cultivation with Clifford measurements by gauging

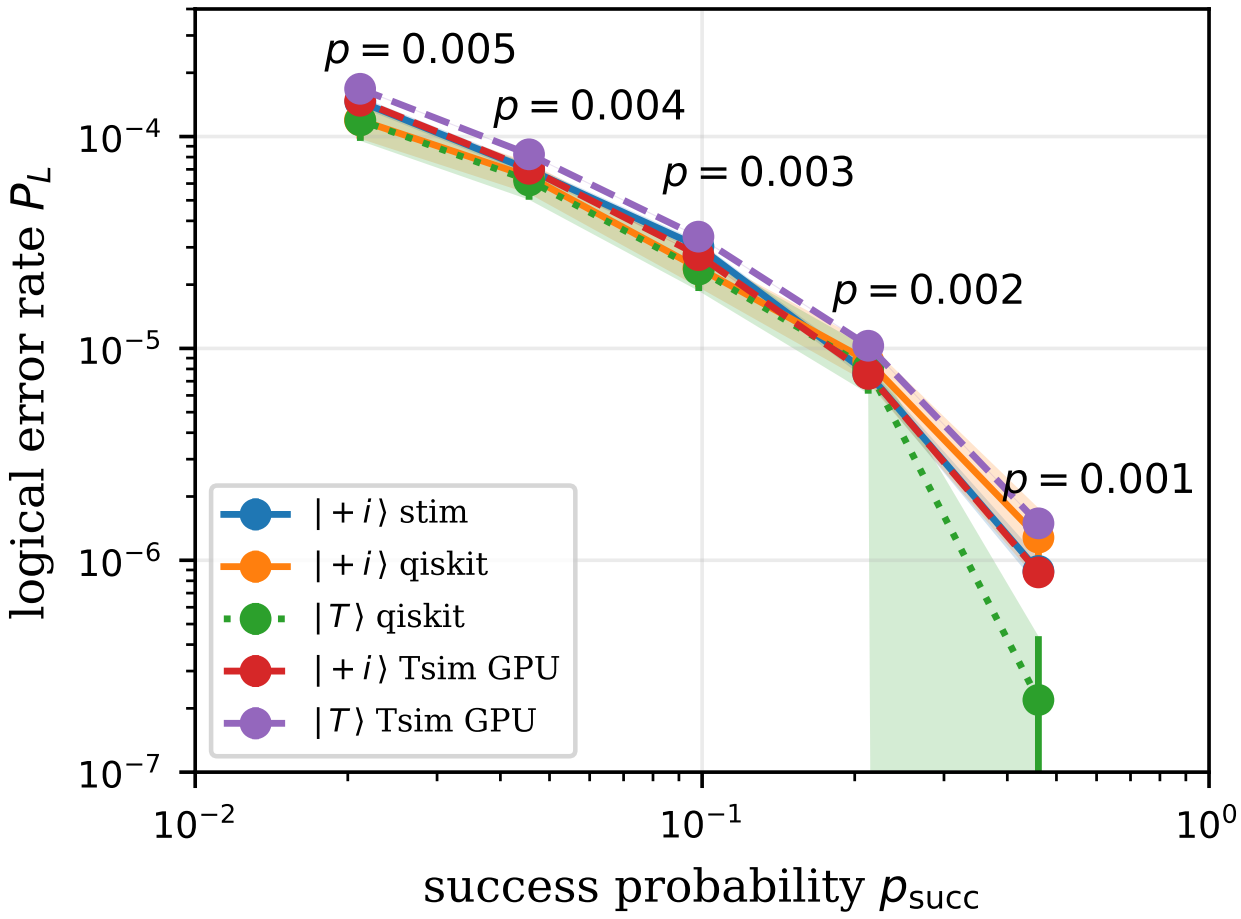
arXiv:2603.05429 Study module `repro/studies/gauging_cultivation.py`

- Fig. 4(a): $d = 3$, $p = 0.1$–0.5%: logical error rate vs. success probability of the authors' Stim simulation of the $|+i\rangle$ proxy, and of qiskit state-vector simulations of $|+i\rangle$ and $|T\rangle$.
- Fig. 4(b): success probability vs. p (1e-4 to 1e-3) for gauging and for Gidney et al.'s cultivation, $d = 3, \ldots, 11$.
- Fig. 4(c) and Table 1: P_L and expected volume for $d = 3, 5$ at $p =$ 1e-3 and 5e-4 (the $d = 7$ entries are skipped here, as agreed).

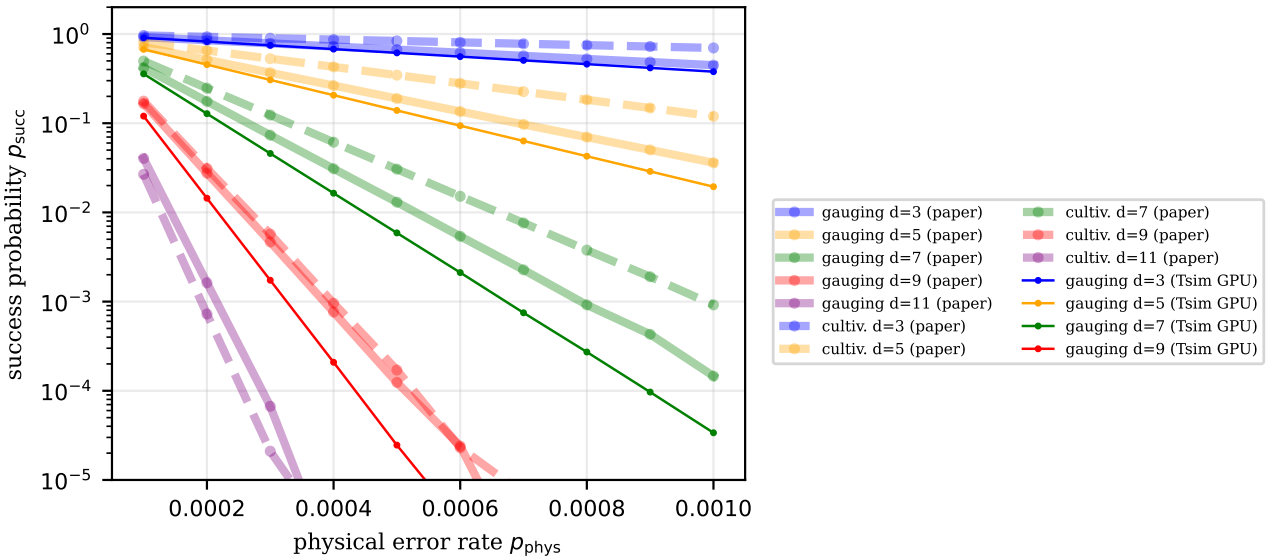
Every detector is postselected, including a final noiseless stabilizer round.

Circuits: the authors' generator `get_stim_circ_w_growing(d, basis="Y", pipeline=True)` with their `utils.add_noise` (uniform p), as in their notebook (`gauging_clifford_measurement.ipynb`, IBM at ce90c09). Clifford proxy: as generated, final noiseless Y_L readout. T version: every $S/S^\dagger$ becomes $T/T^\dagger$ (as in the authors' qiskit check), and a noiseless logical rotation $\exp(\mp i\pi Z_L/8)$ (`repro/magic_check.py`) is inserted before the readout so that Y_L checks the T state deterministically, the same observable as the paper's per-shot $XS^\dagger$ expectation. For Fig. 4(b), exactly the circuits of the notebook's yield comparison: `pipeline=False` for gauging, and the repository's `cultivation_stim_circuits/noiseless_cultivation_circuit_d*.stim` with `add_noise` for cultivation (success probability only, Clifford proxies).

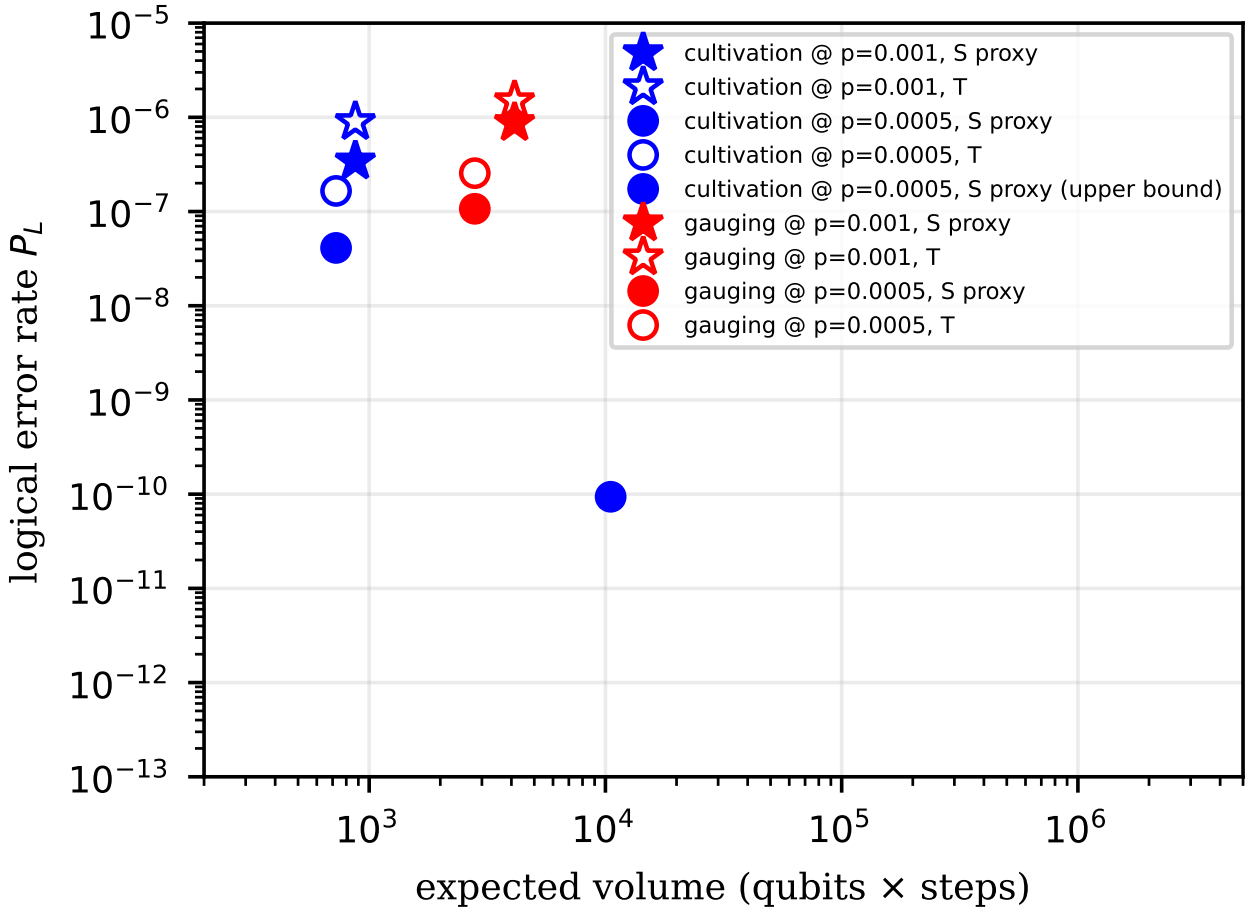
Figures (a) and (b) are drawn with the notebook's own plotting code (cells 18 and 7) and its data (cell 17; `yield_comparison/*.pkl`) overlaid.



Hetényi et al. Fig. 4(a): $d = 3$, logical error rate vs. success probability
Compute: 3.1 min GPU sampling + 4.2 s compile, 5.4e11 shots, 12 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Drawn with the authors' notebook code (cell 18) and data (cell 17): $|+i\rangle$ stim, and their qiskit state-vector $|+i\rangle$ and $|T\rangle$; dashed: ours on the GPU ($|+i\rangle$ proxy and real T, p = 1e-3 to 5e-3).



Hetényi et al. Fig. 4(b): success probability, gauging vs. cultivation, $d = 3, \ldots, 11$
Compute: 4.4 s GPU sampling + 46 s compile, 4e9 shots, 40 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
The notebook's cell 7 on its pickled data (thick, faint) and ours on the GPU (thin) for exactly the notebook's circuits (non-pipelined gauging; the repository's cultivation circuits with `add_noise`); 10^8 shots per point.



Hetényi et al. Fig. 4(c): logical error rate vs. expected volume, $d = 3, 5$
Compute: 33.1 min GPU sampling + 32 s compile, 9.3e12 shots, 14 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Stars: $p = 10^{-3}$; circles: $p = 5 \times 10^{-4}$; filled: Clifford proxy; open: real T. Volume = qubits × TICK layers of the circuit, divided by the success probability (the paper's exact volume convention is not released, so the horizontal positions may be shifted by a constant factor). Cultivation points are the Gidney study's.

Hetényi et al. Table 1 (the $d_f = 7$ rows are not attempted)
Logical error probability P_L and success rate p_{succ}; Tsim errors in parentheses.
Compute: 33.1 min GPU sampling + 32 s compile, 9.3e12 shots, 14 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

		paper ($\|+i\rangle$, stim)		Tsim GPU, $\|+i\rangle$		Tsim GPU, T	
p_{phys}	d_f	P_L	p_{succ}	P_L	p_{succ}	P_L	p_{succ}
0.1%	3	$(8.8 \pm 0.8) \times 10^{-7}$	46%	8.8e-07 ± 6e-09 (20043)	46.1%	1.5e-06 ± 1e-08 (20180)	46.1%
	5	$(3.2 \pm 0.4) \times 10^{-10}$	4%	–	–	–	–
0.05%	3	$(1.2 \pm 0.2) \times 10^{-7}$	65%	1.1e-07 ± 8e-10 (20004)	67.9%	2.6e-07 ± 2e-09 (20030)	67.9%
	5	$(1.1 \pm 0.3) \times 10^{-11}$	20%	–	–	–	–

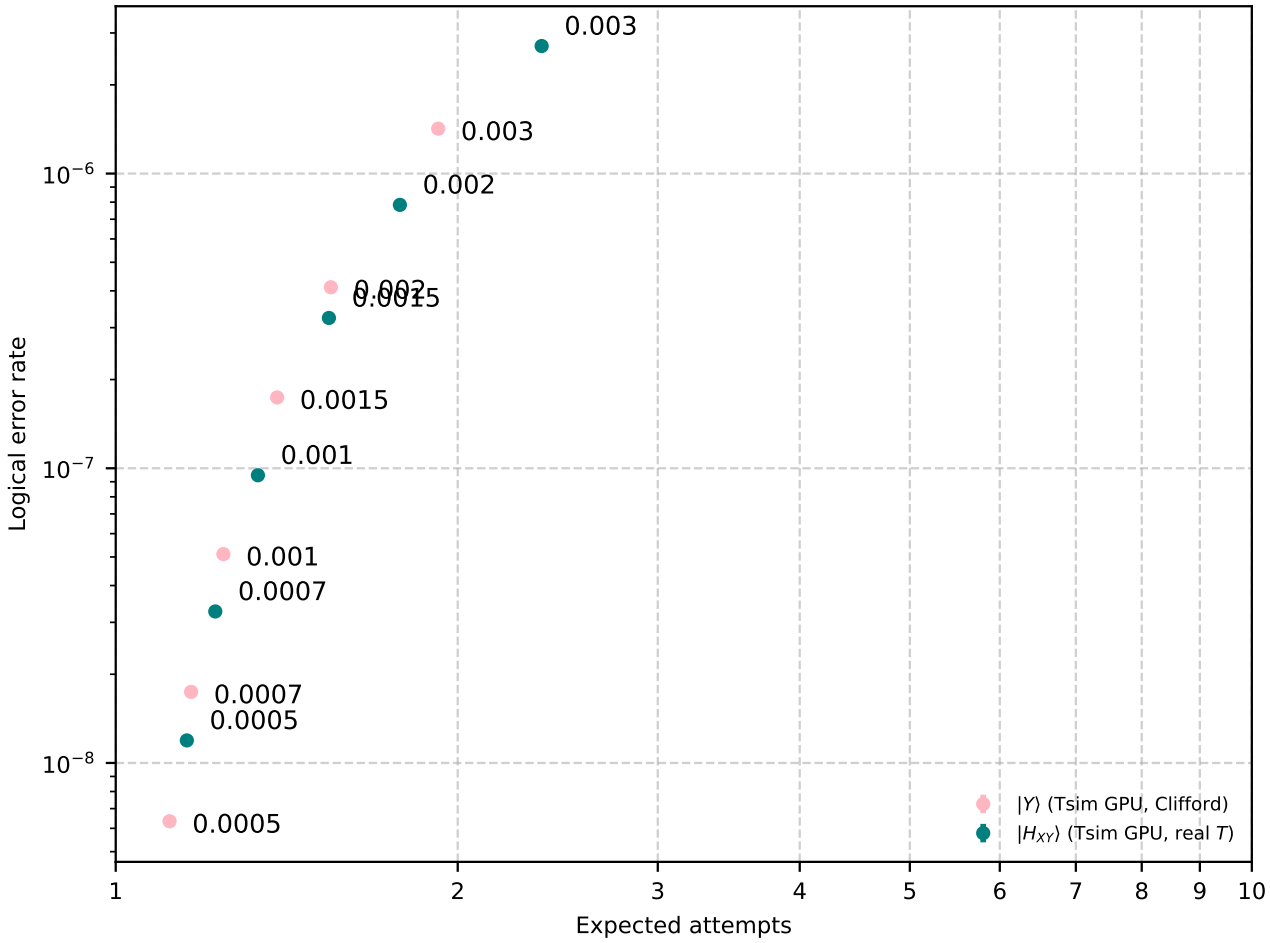
6 Sahay et al. (2025), Fold-transversal surface code cultivation

arXiv:2509.05212 Study module `repro/studies/fold_transversal.py`

The paper compares $|Y\rangle$-state cultivation (Clifford, Stim) with the true $|H_{XY}\rangle = T|+\rangle$ cultivation (a state-vector simulation) on the fault-distance-3 surface-code protocol: unitary preparation plus stabilizer round, two fold-transversal H_{XY} checks with GHZ(3) ancillas, then a noiseless decoding of the code and a logical readout; every detector postselected.

Circuits:

- Clifford proxy: the authors' `generate_sv_cirquit(p, dfinal=7, prep="unitstab", cirq=False)` (kaavyas99/MSC_ at 9378fd2), the circuit their state-vector handoff starts from; noiseless logical Y readout.
- T: the native-T reconstruction of the same circuit by Wan and Zapirain (arXiv:2609.18922, `fold-d3/model.json`: CY layers -> CCZ + controlled-SX), exported by our `make_circuits.py` at noise p, followed by a noiseless $T^\dagger$ and X readout of the decoded logical qubit.



Sahay et al. Fig. app_hxy(b) (`ycult_v2.pdf`): $|Y\rangle$ vs. $|H_{XY}\rangle$ cultivation, fault distance 3, no escape
Compute: 46.4 min GPU sampling + 9.7 s compile, 3.7e12 shots, 12 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same axes and style as the paper (logical error rate vs. expected attempts, $\pm 1\sigma$ bars, points labelled by p; the paper shows $p = 0.001, 0.003, 0.005, 0.007$). All detectors postselected; noiseless decoding and logical readout. No numbers for this panel are released (the repository's `all_paper_data_v2.csv` holds only end-to-end runs), so there is no overlay. Read off the paper's figure, its $|Y\rangle$ stabilizer point at $p = 10^{-3}$ is about 1.8×10^{-7} at 1.35 attempts, versus 5×10^{-8} at 1.25 here, so the paper's circuit for this panel has more fault locations than `generate_sv_cirquit(prep="unitstab")`; as in the paper, $|H_{XY}\rangle$ and $|Y\rangle$ differ once escape is excluded (here H_{XY} is about 2× worse, in the paper about 2× better).

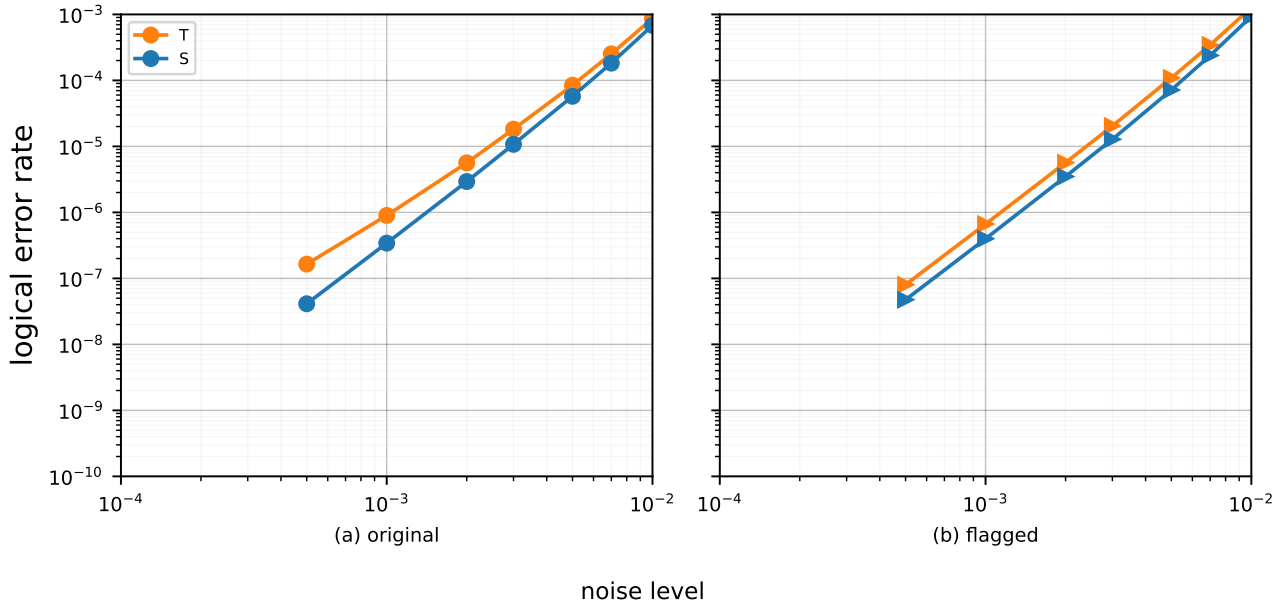
7 Chan, Strikis, Sun, Cai (2026), Diagnosing and restoring the degraded fault distance of magic state cultivation

arXiv:2609.17706 Study module `repro/studies/chan_flags.py`

Pre-escape colour-code cultivation (injection + double checks, every detector postselected, noiseless final check), with and without the paper's Z flags in the final double check:

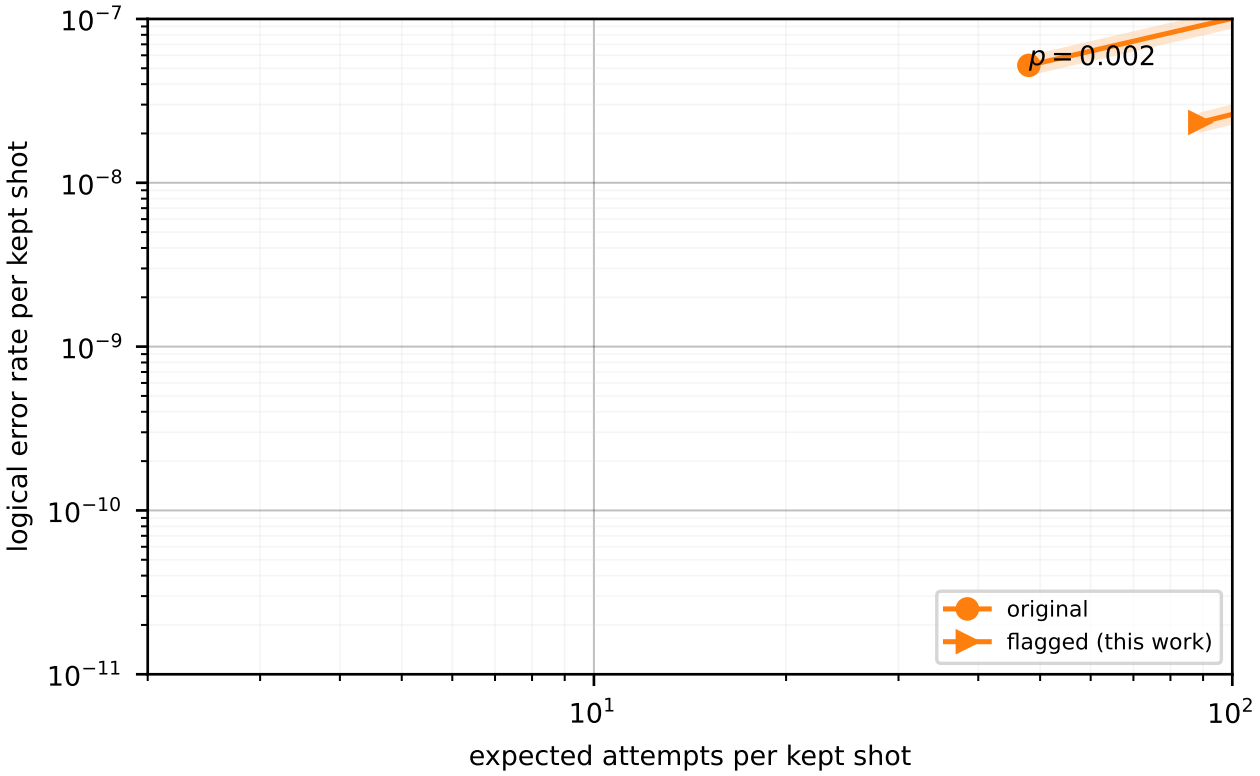
- Fig. `d3_pre_escape_ler`: $d = 3$ ($a = 6$ ancillas, $f = 2$ flags), S vs. T;
- Figs. `d5_pre_escape_ler` and `improvement_and_cost_ratios`: $d = 5$ ($a = 19$, $f = 13$), T state, original vs. flagged. The abstract quotes a 4.9x LER improvement at 1.36x attempts for $p = 10^{-3}$, and 10.9x at 1.17x for $p = 5 \cdot 10^{-4}$.

Circuits: the package's bundled S-state references (timchan0/cliffordea at 27385c6, `sim/stim_files`, the $d = 5$ ones with the Bell-growth feedforward and detector healing) and its own `make_variant_text`: `replace_noise_level` from 10^{-3} to p, and `swap_t_and_s_gates` for the T state. The paper sampled with SymFT.

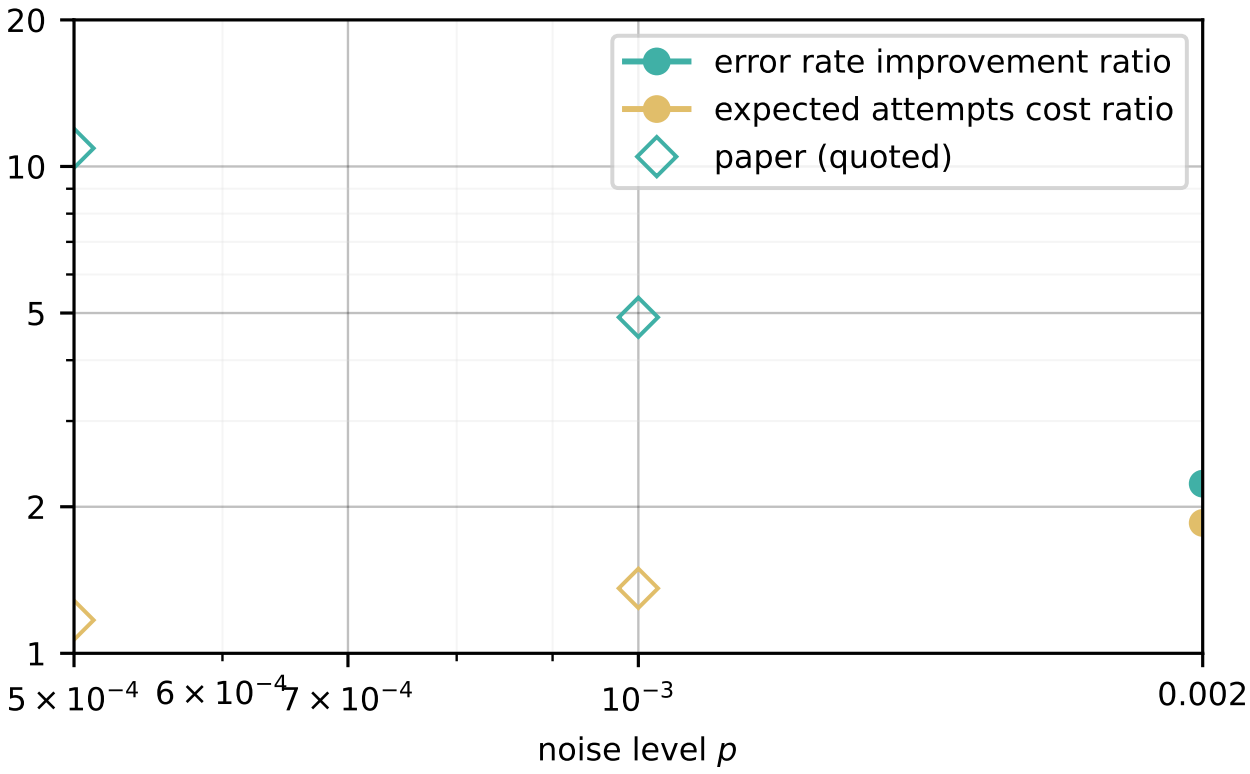


Chan et al. Fig. d3_pre_escape_ler: $d = 3$ cultivation, S vs. T, (a) original and (b) Z-flagged double check

Compute: 19.9 min GPU sampling + 11 s compile, 1.2e12 shots, 28 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same layout as the paper (sinter's `plot_error_rate`; shaded: 1000x likelihood). Orange: real T gates; blue: the S proxy. With the flags the T-state error recovers fault distance 3. The paper sampled with SymFT and released no numbers.



Chan et al. Fig. d5_pre_escape_ler (`summary.pdf`, right): $d = 5$ T **cultivation, original vs. Z-flagged**
Compute: 21.5 min GPU sampling + 14 s compile, 9.7e12 shots, 4 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same axes as the paper; points at p = 5e-4, 7e-4, 1e-3, 2e-3 (and 3e-3).



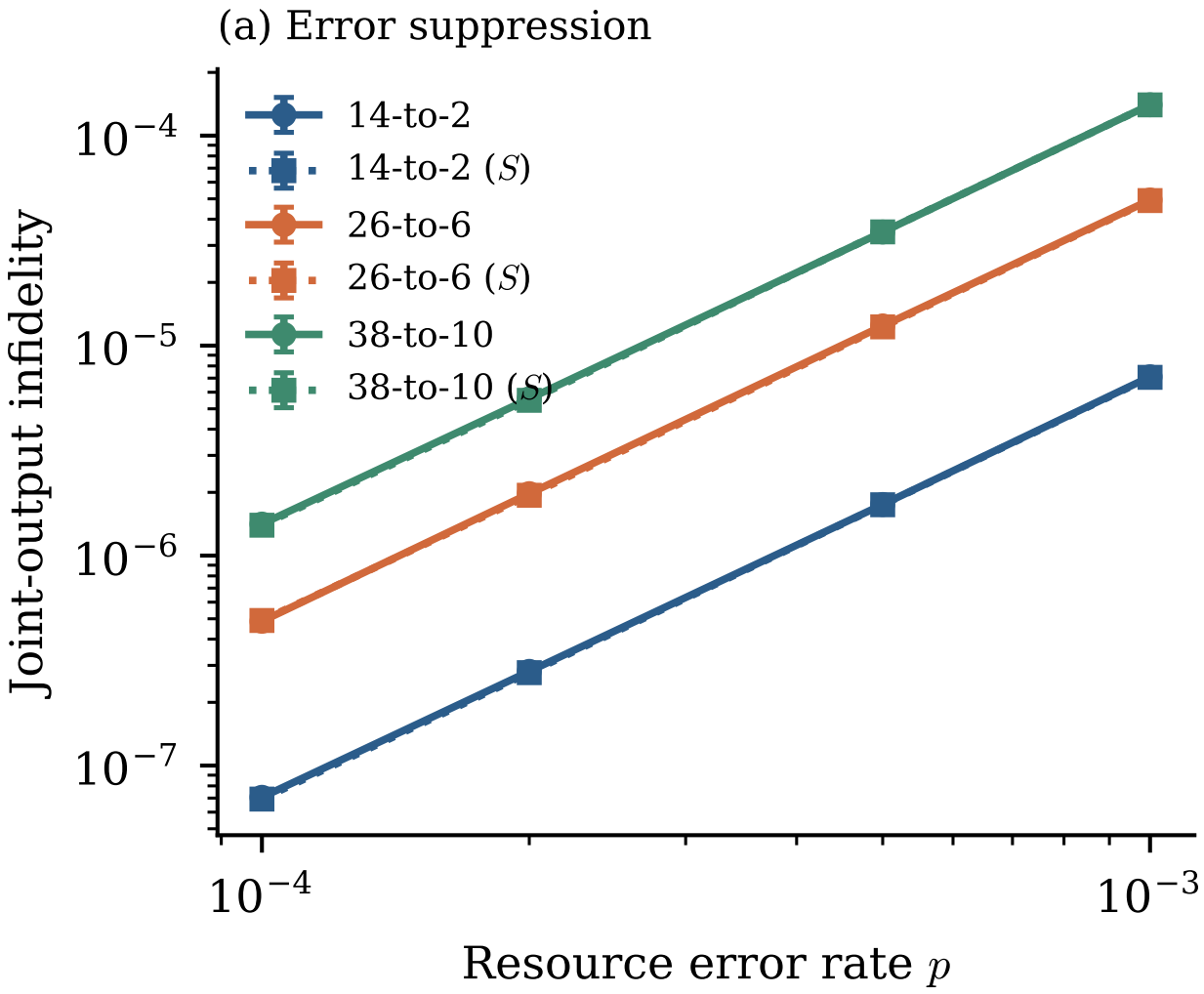
Chan et al. Fig. improvement_and_cost_ratios: $d = 5$ Z flags, LER improvement and attempts cost
Compute: 21.5 min GPU sampling + 14 s compile, 9.7e12 shots, 4 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Improvement = original/flagged logical error rate (median and 95% interval of a Beta-posterior bootstrap); cost = flagged/original expected attempts. Open diamonds: the ratios quoted in the paper's abstract and Sec. VI (4.9x at 1.36x for $p = 10^{-3}$; 10.9x at 1.17x for $p = 5 \times 10^{-4}$).

8 Wang, Gu (2026), Magna: simulating magic state factories as cheaply as Clifford circuits

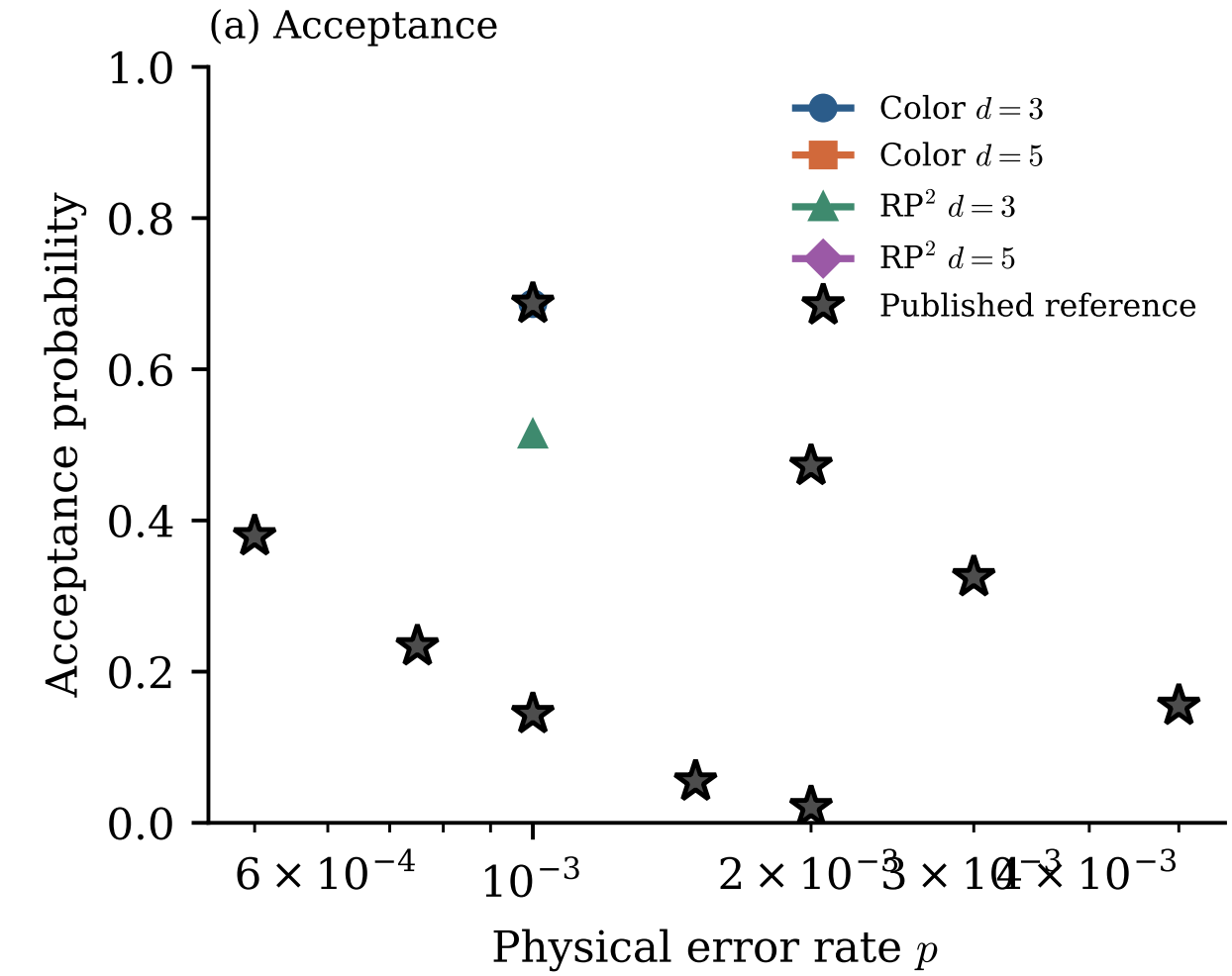
arXiv:2610.22315 Study module `repro/studies/magna.py`

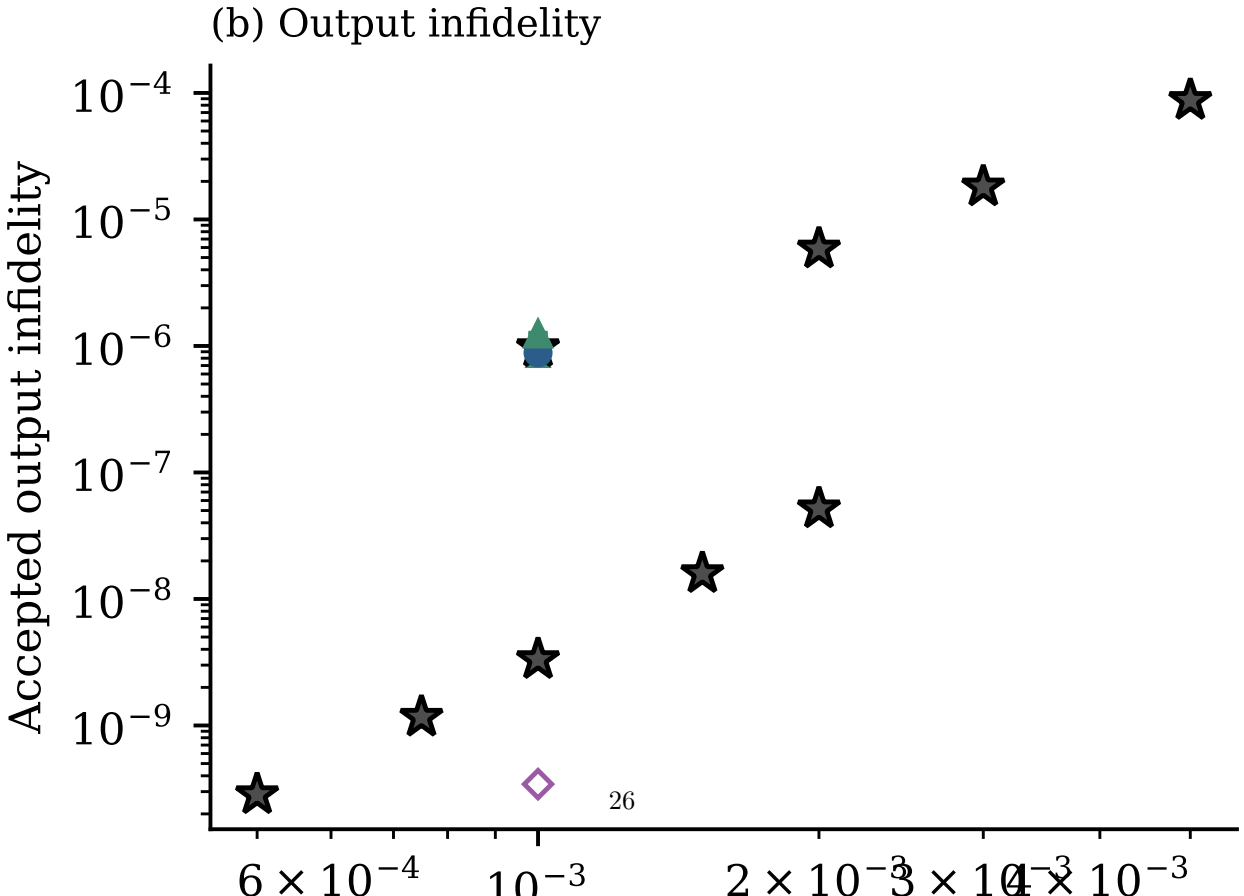
- Fig. 8(a): Bravyi–Haah $(3k+8)$-to-k distillation on bare qubits, $k = 2, 6, 10$. Every input magic state has a Z error with probability p; everything else is noiseless. Plotted: the joint infidelity of the k outputs with $|T\rangle^{\otimes k}$, for p from 10^{-4} to 10^{-3}.
- Fig. 9(a,b): colour-code and RP^2 cultivation at $d = 3, 5$ (no escape), acceptance rate and infidelity of accepted runs under uniform depolarizing noise.

Circuits: Magna's own example protocols (itpyi/magna at 01ae9c2, `examples/*/physical.magna`), translated to the Tsim dialect by our `make_circuits.py` (deferred-measurement controlled corrections etc.). The examples are written at $p = 10^{-3}$ and Magna ships no generator, so other noise strengths are made by changing that literal. Only the detectors named in the examples' `POSTSELECT` lines are postselected. The error check is the examples' task definition: undo the ideal target circuit on the output qubits and require every output to read 0 (a joint check). Clifford versions: Magna's `-S` examples (product-S targets); for the colour code the S proxy is in the Gidney study.



Magna Fig. 8(a): Bravyi–Haah $(3k+8)$-to-k distillation, error suppression
Compute: 7.4 min GPU sampling + 8.2 s compile, 9e11 shots, 24 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same axes and style as the paper. Bare qubits, Z errors of probability p on the input magic states only; bars: 1.96σ. Solid: real T inputs; dotted: the matched Clifford circuit ($|S\rangle$ inputs, product-S target, which the paper uses for timing only). The two coincide, as they should with Z errors only. Magna's own numbers are not released.


(a) Acceptance
Color $d = 3$
Color $d = 5$
RP^2 $d = 3$
RP^2 $d = 5$
Published reference
Acceptance probability
1.0
0.8
0.6
0.4
0.2
0.0
6×10^{-4}
10^{-3}
2×10^{-3}
3×10^{-3}
4×10^{-3}
Physical error rate p


(b) Output infidelity
Accepted output infidelity
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
6×10^{-4}
10^{-3}
2×10^{-3}
3×10^{-3}
4×10^{-3}

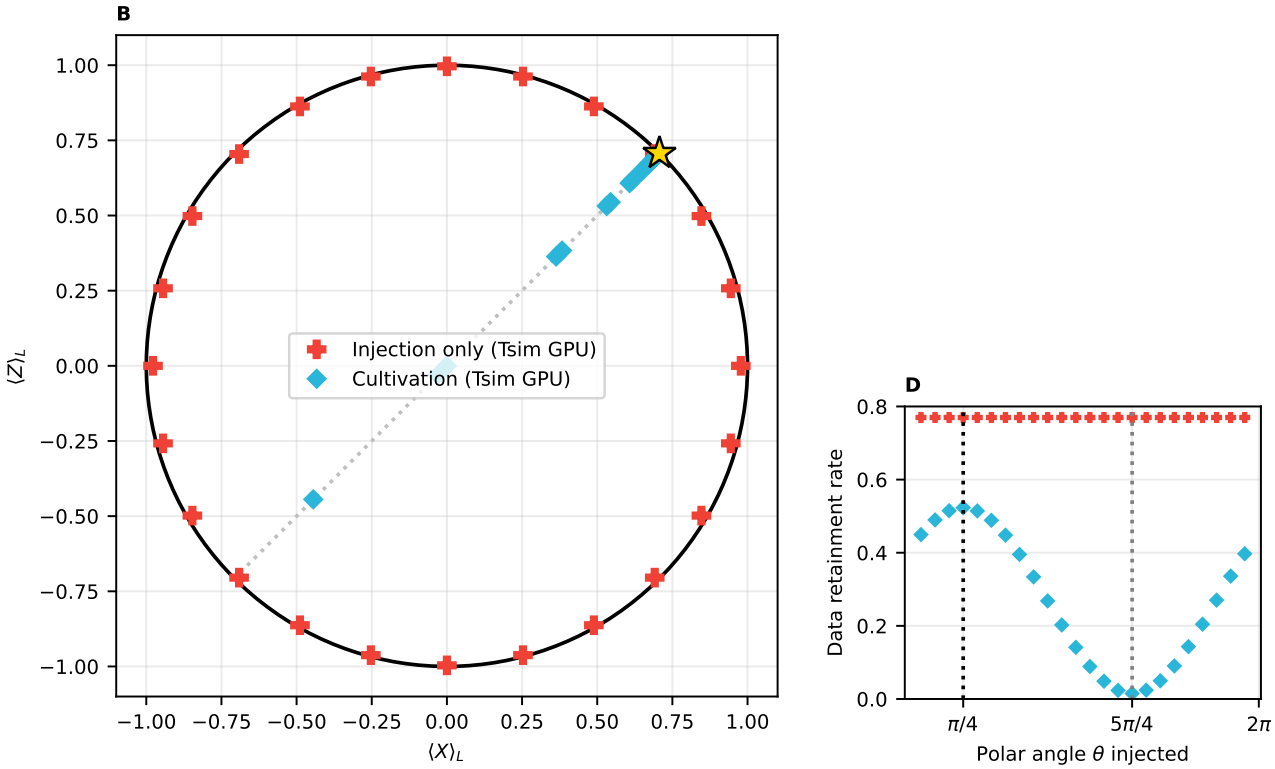
9 Rosenfeld et al. (2025), Magic state cultivation on a superconducting quantum processor

arXiv:2512.13908 Study module `repro/studies/google_experiment.py`

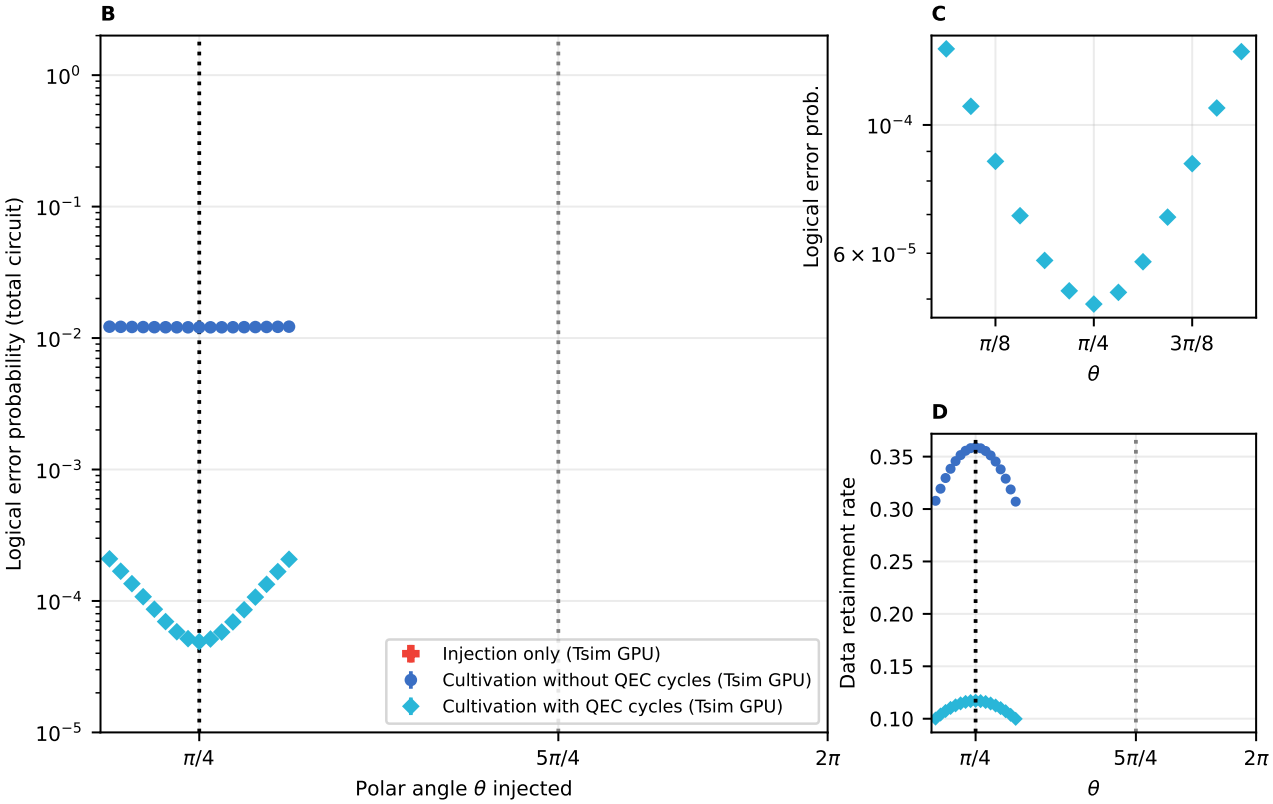
The paper's grey points are state-vector simulations of the experimental circuits under the SI1000 noise model; shots are kept only if no detector fires (post-selected logical tomography). Here:

- Fig. 2(b,c): logical tomography ($\langle X_L\rangle$, $\langle Z_L\rangle$) of injection only and of injection + cultivation, vs. the injected angle θ, SI1000 $p = 3.0 \times 10^{-3}$;
- Fig. 3(b): kickback tomography (KT) of the cultivated $|T\rangle$: logical error vs. θ, with and without the QEC cycles, SI1000 $p = 2.3 \times 10^{-3}$; SI Fig. `validating_KT`: KT error at the T angle vs. p (expected $\propto p^3$);
- Fig. 4(b): tomography after grafting to the surface code, injection only and cultivation with $N = 2, 3$ cycles, SI1000 $p = 2.3 \times 10^{-3}$.

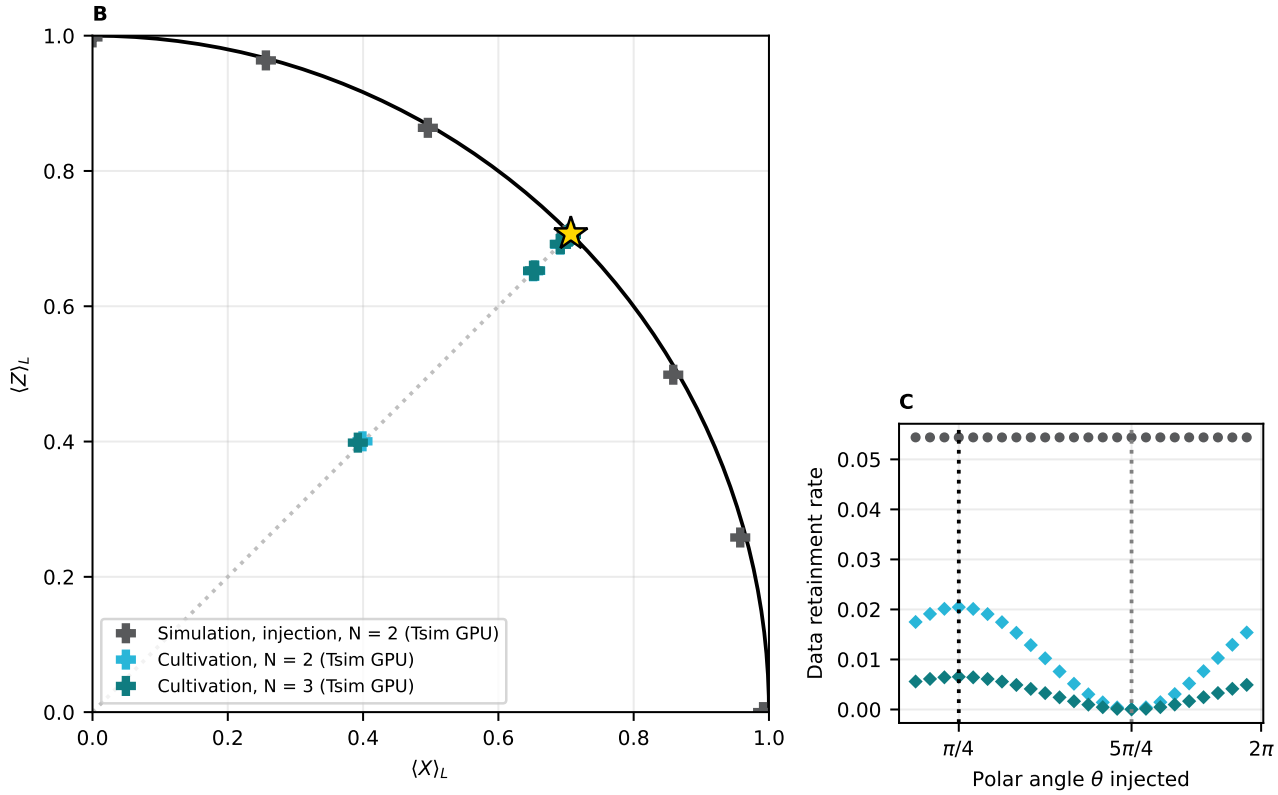
Circuits: the released Zenodo (17872003) circuits, repaired to real gates from their Cirq versions by our `make_circuits.py` (injection $e^{-i\theta Y/2}$ as an H_{YZ}-conjugated Z rotation, $T/T^\dagger$ checks). The injection rotation is set to $R_Z(\theta/\pi)$; SI1000 noise is added with Gidney's `gen.NoiseModel.si1000(p)` (the model the paper cites), with non-Clifford gates mapped to Clifford placeholders while the noise is inserted. The experiment had no Clifford proxy, so there is none here.



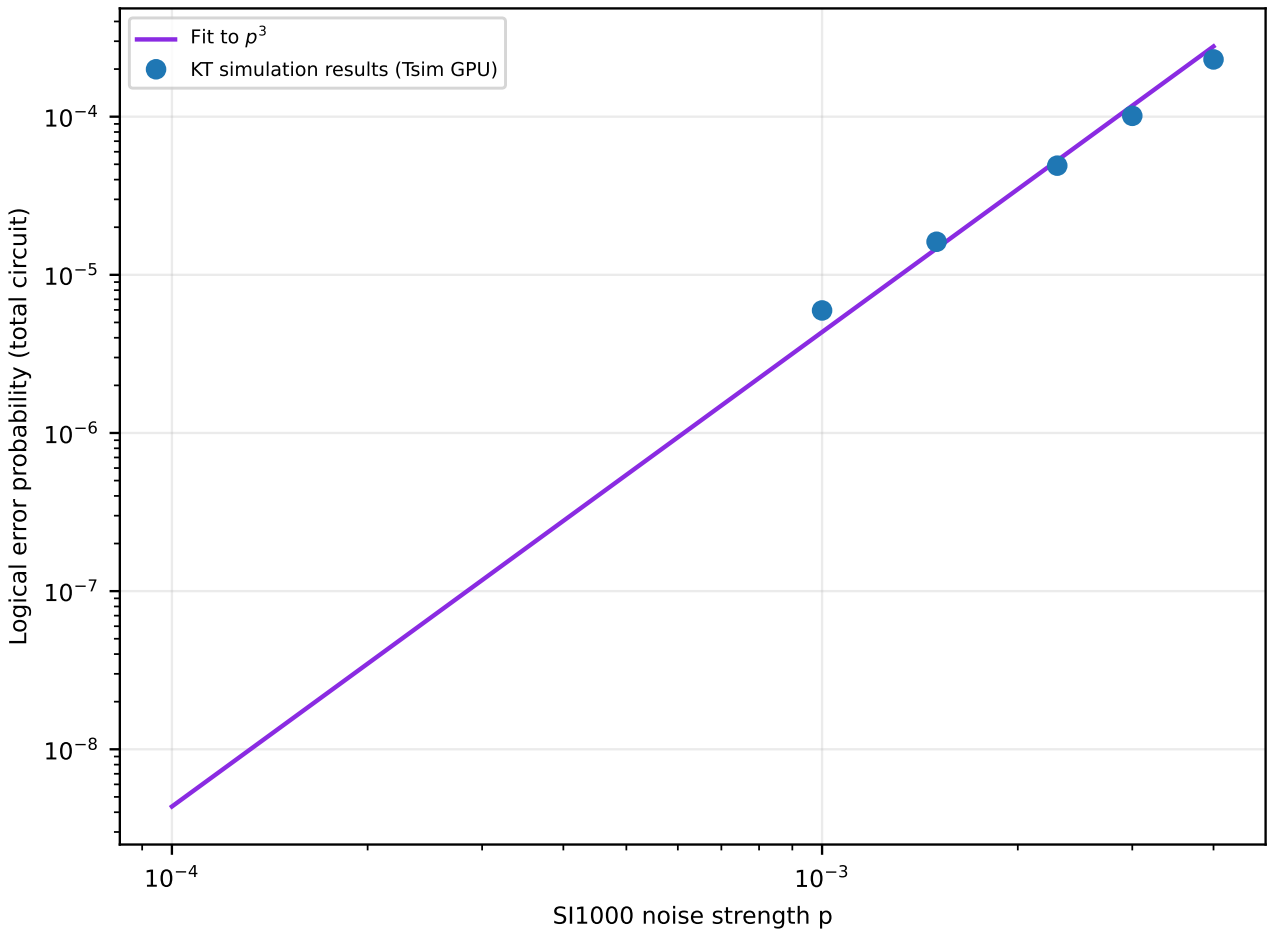
Rosenfeld et al. Fig. 2(B, D): post-selected logical tomography and data retainment vs. injected angle
Compute: 1.4 min GPU sampling + 41 s compile, 2.9e11 shots, 96 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Layout and markers of the paper's Fig. 2 (panels B and D). Our points replace the paper's grey state-vector *simulation* points (SI1000, $p = 3.0 \times 10^{-3}$): red, injection only; cyan, injection + cultivation, which is projected onto the magic axis (dotted); star: the ideal $|T\rangle$. 24 angles, up to 3×10^9 shots per basis. The paper's experimental points are not released in numeric form here.



Rosenfeld et al. Fig. 3(B, C, D): kickback tomography (KT) vs. injected angle
Compute: 2.1 min GPU sampling + 33 s compile, 6.8e11 shots, 34 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Layout and markers of the paper's Fig. 3: logical error of the whole KT circuit vs. θ (B), zoom near $\pi/4$ (C), data retainment (D), SI1000 $p = 2.3 \times 10^{-3}$. All three curves are Tsim GPU simulations of the released circuits (the paper's grey points are its state-vector simulation of the cultivation-with-QEC circuit). Without the QEC cycles there is a single stabilizer measurement at the end, so single faults cause logical errors: the error is about 100x worse (as in the paper) and, since postselection on the cultivation checks still projects onto the T axis, nearly independent of θ.



Rosenfeld et al. Fig. 4(B, C): tomography after grafting to the surface code
Compute: 42 s GPU sampling + 1.6 min compile, 4.3e11 shots, 144 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Layout of the paper's Fig. 4 (positive quadrant of the X–Z plane, data retainment vs. θ), SI1000 $p = 2.3 \times 10^{-3}$, the released injection-only simulation circuit and the cultivation circuits with $N = 2, 3$ cycles.



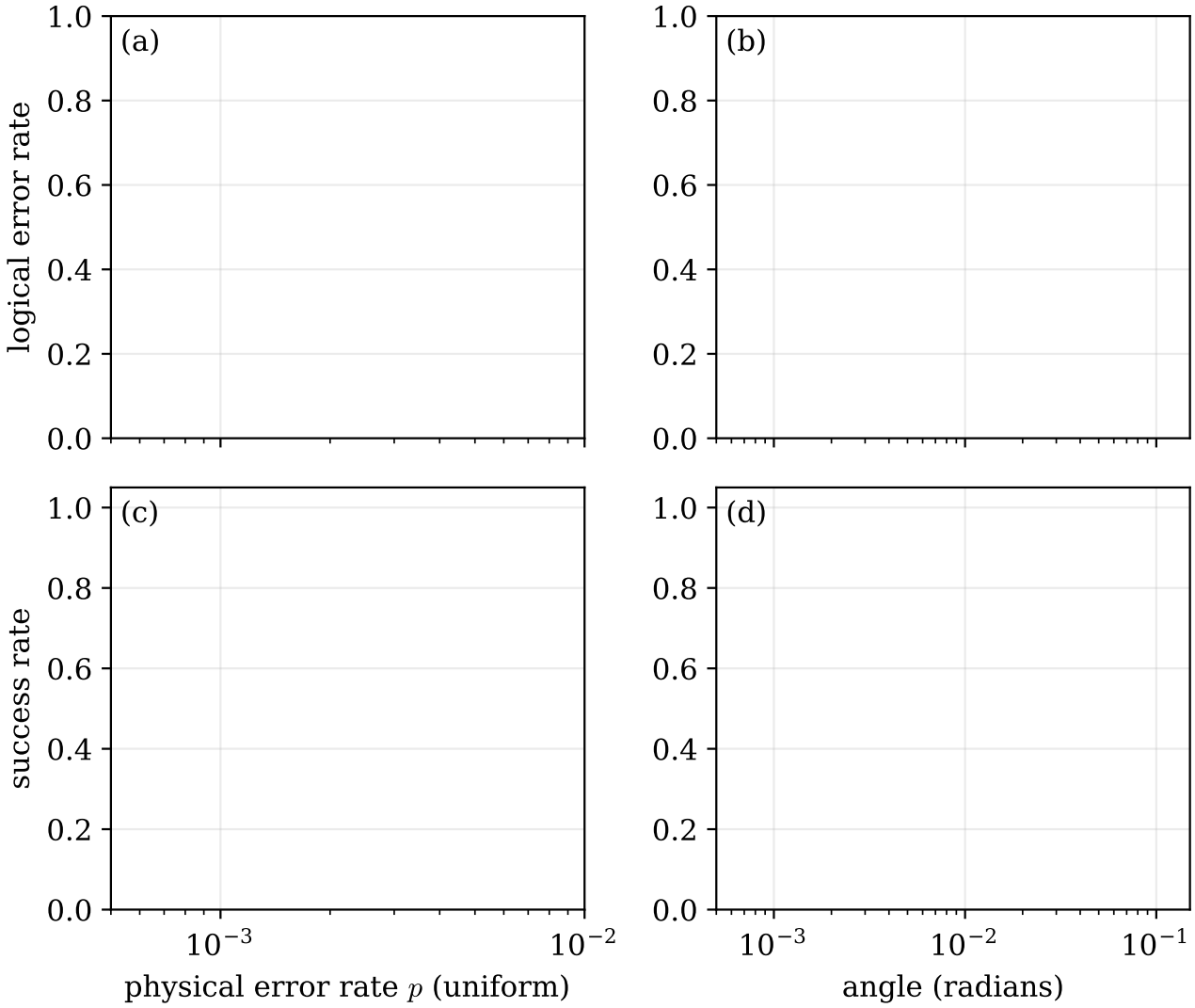
Rosenfeld et al. SI Fig. validating_KT: KT error at the T angle vs. SI1000 noise strength
Compute: 22 s GPU sampling + 2.9 s compile, 1.2e11 shots, 6 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Same axes as the paper (KT of the cultivated $|T\rangle$, fit to p^3), at the paper's noise grid. Our values are somewhat below the paper's (e.g. 2.2×10^{-4} vs. $\sim 5 \times 10^{-4}$ at $p = 4 \times 10^{-3}$), consistent with a slightly different SI1000 variant (we use Gidney's `gen.NoiseModel.si1000`).

10 Ismail et al. (2025), Transversal architecture for megaquop-scale quantum simulation with neutral atoms (transversal STAR)

arXiv:2509.18294 Study module `repro/studies/star_injection.py`

The paper's Fig. 6 shows the logical error rate and success rate of its transversal multi-rotation (TMR) STAR injection with full postselection, vs. the physical error rate (at a logical angle of 10^{-2}) and vs. the angle; Fig. 13 validates against an analytic success rate at $d = 3$.

The authors released no generator; the circuits here are QuEra's public STAR challenge circuits (`QuEra-Challenge/star_cir` with the challenge's own `physical_angle` helper to set the logical angle. They differ from the paper's Fig. 6 protocol: $k = d$ physical rotations (the paper uses $k = 3$; equal at $d = 3$), d postselected rounds after the rotations, uniform depolarizing noise ($p = 10^{-3}$ in the released files; other p by changing that literal), and a noiseless logical un-rotation and X readout. Every detector is postselected (raw parity; the $d - 1$ detectors right after the rotations are random by design). The paper's Clifford approximation is not specified, so no Clifford version is sampled; the noiseless success probability $\cos^{2k}(\phi/2) + \sin^{2k}(\phi/2)$ of k physical rotations by ϕ is overlaid.



Ismail et al. Fig. 6: STAR small-angle injection with full postselection (challenge circuits)
Compute: not yet sampled
The paper's layout and grid: (a, c) logical error and success rate vs. error rate at a logical angle of 10^{-2} rad; (b, d) vs. the logical angle at $p = 7.3 \times 10^{-4}$. Same colours ($d = 3, 5, 7$) and markers as the paper. The protocol is that of QuEra's public STAR challenge circuits ($k = d$ rotations, d postselected rounds after them, uniform depolarizing noise p on every gate instead of the paper's hardware-aware model whose x-axis is the total two-qubit error rate), so the curves are not expected to coincide with the paper's; the authors released no generator for the paper's circuits.

11 Chen et al. (2025), Efficient magic state cultivation on RP^2

arXiv:2503.18657 Study module `repro/studies/rp2_ungrown.py`

Table 1 lists, at $p = 10^{-3}$, the logical error rate and discard rate of each cultivation process alone (no escape stage), benchmarked by sampling the Clifford variant (S for T, cultivating $S|+\rangle$) with a final round of perfect stabilizer measurement and logical readout: MSC-3 8.3×10^{-7} / 48.6\%, MSC-5 3.6×10^{-10} / 90.5\%.

Circuits: the authors' `circuit_garage` files (Zihan-Chen-PhMA/Cultiv_T_RP2 at 77ad9b3, written at $p = 10^{-3}$ by their `d3_T_cult_sampling.py` / `d5_T_cult_sampling.py`). Clifford version: as released. T version: $S \to T$, $S^\dagger \to T^\dagger$ (reversing the authors' substitution) plus a noiseless logical rotation (`repro/magic_check.py`) so that the final Y_L readout checks the T state deterministically. MSC-5 with T gates has Tsim active width 10, which makes it the slowest circuit here.

Chen et al. Table 1, ungrown columns: cultivation without escape, $p = 10^{-3}$

The paper's Table 1 rows and ungrown columns (the end-to-end columns need a decoder). MSC-3/5: the authors' RP^2 circuits; MSC ($d_{\text{color}} = 3, 5$): Gidney et al.'s colour-code cultivation, whose Tsim GPU values are the Gidney study's points. Every detector postselected, perfect final stabilizer round and logical readout.

Compute: 23 s GPU sampling + 0.5 s compile, 7.9e10 shots, 2 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)

MSC protocol	paper (S proxy, Stim)		Tsim GPU, S proxy		Tsim GPU, T	
	Logical Error rate (Ungrown)	Discard Rate (Ungrown)	Logical Error rate (errors)	Discard Rate	Logical Error rate (errors)	Discard Rate
MSC-3 (this work)	8.3×10^{-7}	48.6%	8.3e-07 (20095)	48.6%	1.3e-06 (20161)	48.6%
MSC ($d_{\text{color}} = 3$) [Gidney et al.]	3.4×10^{-7}	31.3%	3.4e-07 (20079)	31.3%	9e-07 (20147)	31.3%
MSC-5 (this work)	3.6×10^{-10}	90.5%	–	–	–	–
MSC ($d_{\text{color}} = 5$) [Gidney et al.]	3.5×10^{-10}	85.6%	–	–	–	–

12 Jacoby et al. (2025), Magic state injection with erasure qubits

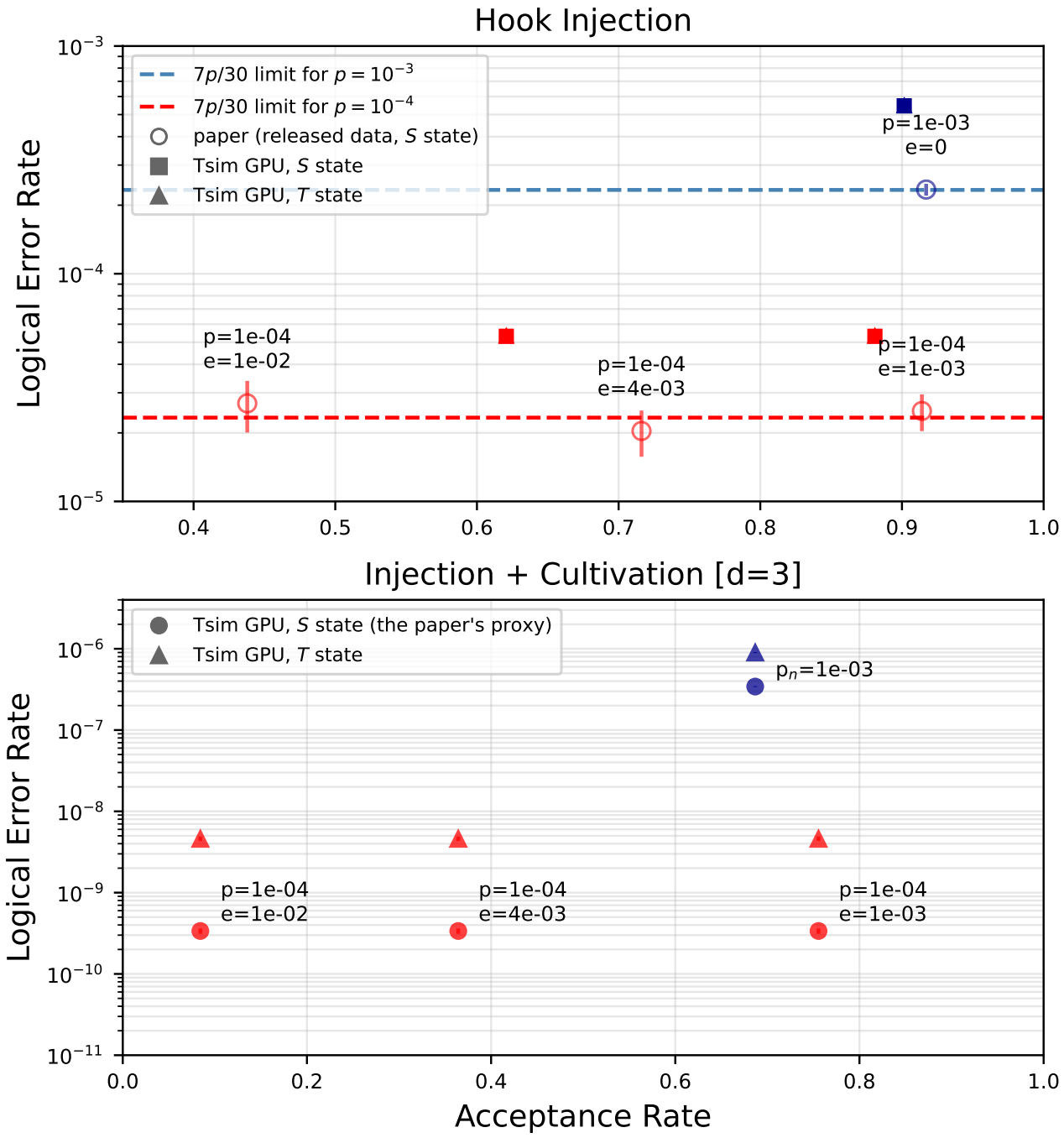
arXiv:2504.02935 Study module `repro/studies/erasure_injection.py`

With erasure qubits, every heralded erasure in the (ungrown, fully postselected) patch discards the shot, so the logical error rate of kept shots depends only on the residual Pauli rate p, and the acceptance rate is the Pauli acceptance times the probability of no erasure. The paper's scenarios: $(e, p) = (0, 10^{-3})$, and $e \in \{10^{-3}, 4 \cdot 10^{-3}, 10^{-2}\}$ with $p = 10^{-4}$; erasure probability e per two-qubit gate and SPAM operation and $e/10$ per single-qubit gate (`end2end_script.py`: `e1=e/10, e_SPAM=e`).

Circuits:

- Fig. 5(b), unitary injection + $d = 3$ colour-code cultivation: Gidney's generator with uniform depolarizing noise (the paper's circuit is Gidney's `inject[unitary]+cultivate` file), S proxy and T; the $e = 0$ point is the Gidney study's $p = 10^{-3}$ point.
- Fig. 2, $d_1 = 3$ hook injection of $|S\rangle$: the released `circuits/QEC_hook_d=3.stim` ($p = 10^{-3}$; $p = 10^{-4}$ by scaling its noise literals, since the notebook that generated the paper's circuits is not released). T version: the injection $S \to T$ and a noiseless logical rotation (`repro/magic_check.py`) so the final Y_L readout checks the T state. Lao–Criger injection and the hybrid layouts are not reproduced.

The no-erasure probability is computed exactly from the Pauli circuit's noise locations, following `get_erased_circuit`.



Jacoby et al. Figs. 2 (hook-injection panel) and 5(b): erasure qubits, ungrown injection and cultivation
Compute: 17.0 min GPU sampling + 1.9 s compile, 2.2e12 shots, 6 points (NVIDIA GeForce RTX 5090, Tsim a2af975f)
Layout of the authors' notebooks (`compare_injections.ipynb`, `cultivation.ipynb`): logical error rate vs. acceptance, blue $p = 10^{-3}$ without erasures, red residual $p = 10^{-4}$ with erasure rate e; bars 1.96σ. Top: open circles are the paper's released data (`out/2024-12-24-compare_injections_5e6runs_2erasures`); our points use the released `QEC_hook_d=3.stim` example, whose noise model differs from the circuits behind the data (its LER is about 2x higher). Bottom: Gidney's `inject[unitary]+cultivate` circuit as in `cultivation.ipynb`; the paper's values (read off the figure: 3.3×10^{-7} at 0.69 for $p = 10^{-3}$) match our S points. Our acceptances include the exact no-erasure probability. The Lao–Criger panel and the hybrid layouts are not reproduced.

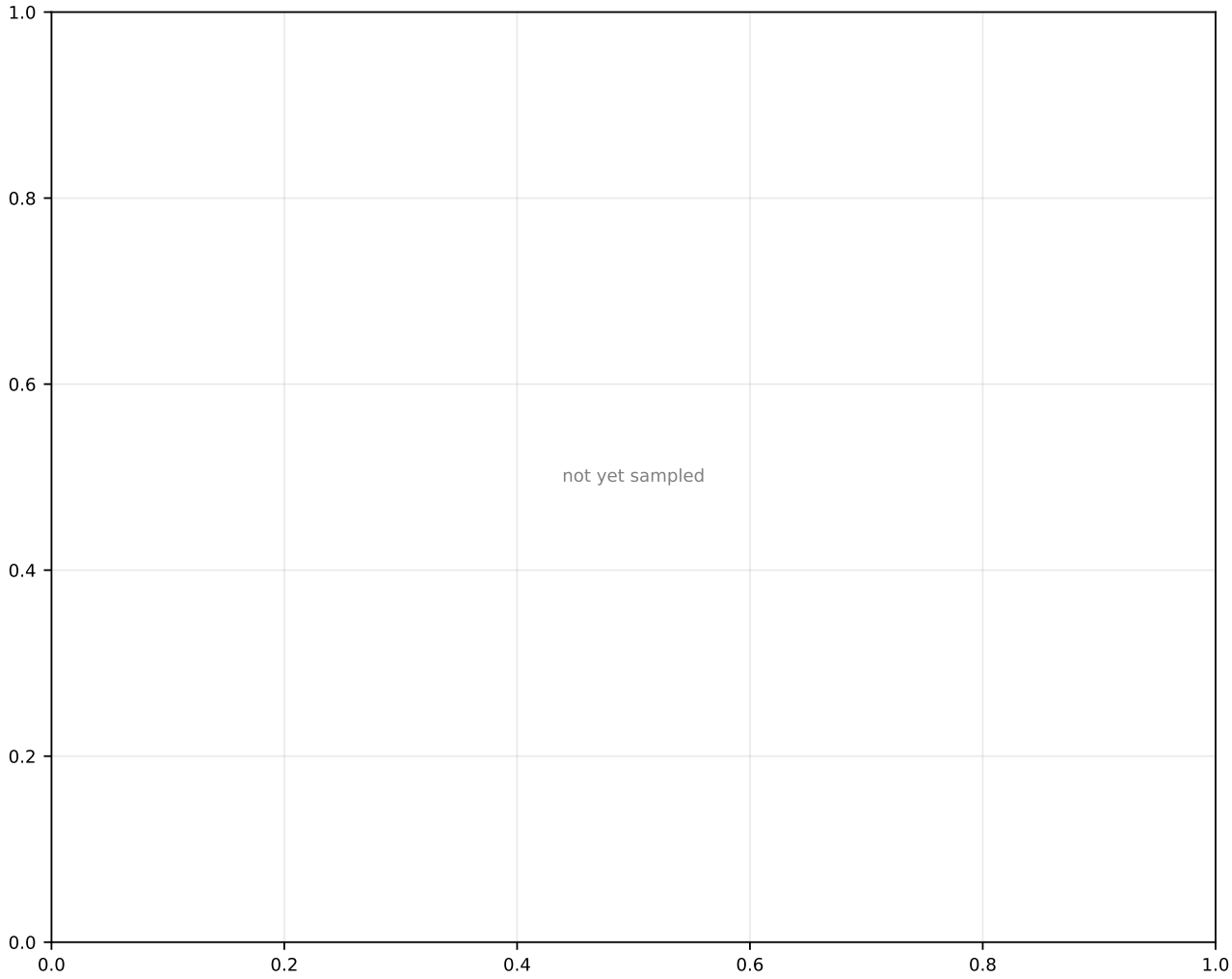
13 Vaknin et al. (2025), Efficient magic state cultivation on the surface code

arXiv:2502.01743 Study module `repro/studies/vaknin_full_vector.py`

Unrotated $d_1 = 3$ surface-code H-type cultivation (injection + double-checking with GHZ ancillas), without the expansion step: every stabilizer measurement postselected, then a perfect logical measurement in the magic-state basis. The authors spent about 10^6 CPU-hours on a full state-vector simulation, roughly 10^7 shots per point, and compared it with their Clifford approximation.

Circuits: the authors' generator (tomirendo/surface-code-magic-state-cultivation at ec28f6b, `Simulation(d=3)`: `generate_syndrome_circuit(detect_x=False)` + `generate_injection_circuit` + `generate_syndrome_circuit` + `generate_double_ghz`, the `injection-syndrome-ghz` variant, whose acceptance matches the paper's Fig. 26) in its own pinned environment (`.venvs/vaknin`, `repro/external/vaknin_generate.py`), with their `NoiseModel().uniform_d`

- Clifford approximation: `to_stim_circuit(apply_non_cliffords=False, non_clifford_noise_strategy="CZ")`, identical to the authors' released `init-H-injection-syndrome-ghz` circuit at $p = 10^{-3}$ (CH checks replaced by CZ, which measures Z_L; their noise kept), final Z_L readout;
- non-Clifford: `apply_non_cliffords=True`, placeholders replaced by $T/T^\dagger$/CCZ/CH/CSWAP decompositions (our `make_circuits.py`); the final noiseless readout is set to Z_L, preceded by the noiseless logical rotation $\exp(\mp i\pi Y_L/8)$ (`repro/magic_check.py`) that turns the final Z_L readout into the magic-basis (H_L) check.



Vaknin et al. Fig. 26
Compute: not yet sampled